

Molecular dynamics

General concepts

Simon Gravelle
LIPhy (Grenoble France), CNRS



What you will learn today

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- The very basics of molecular dynamics

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To go further:

Understanding Molecular Simulation by Daan Frenkel and Berend Smit

<https://lammptutorials.github.io>

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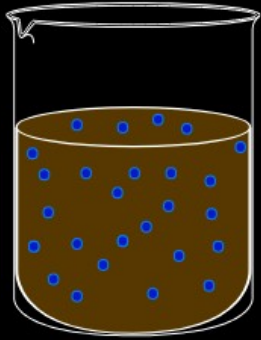
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- How to launch and modify a simple input

Why use molecular dynamics (MD) ?

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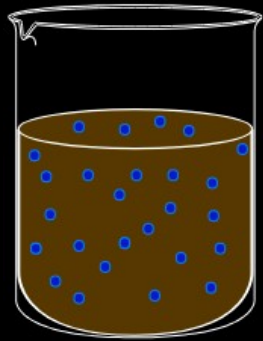
Experiment



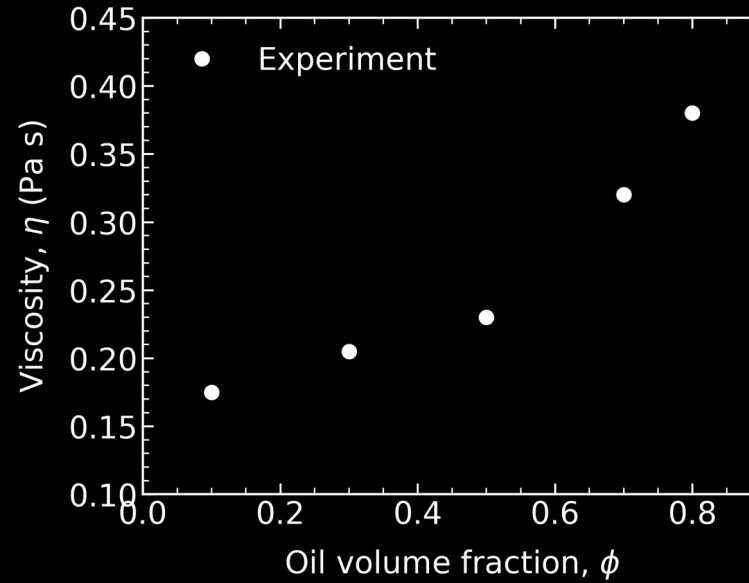
Oil-in-water
Emulsion

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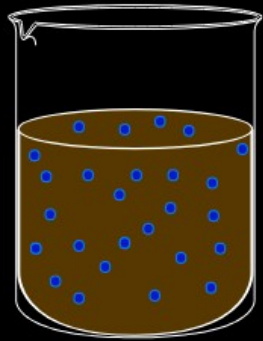


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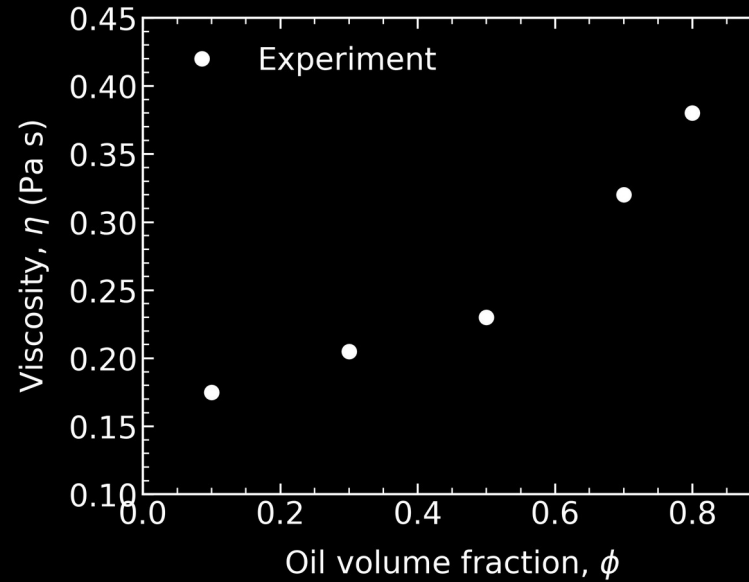


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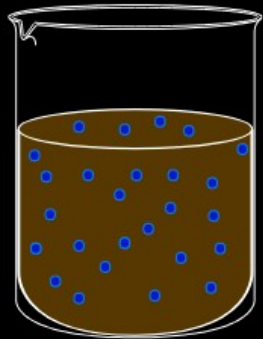
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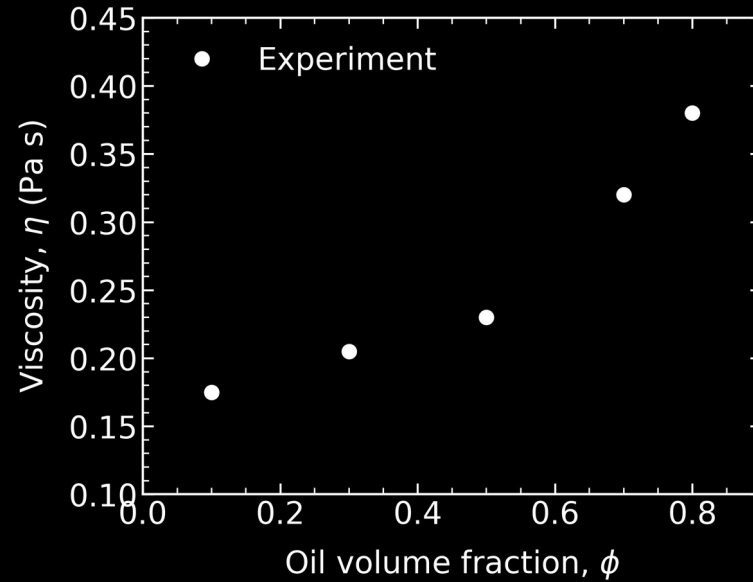
Experimental results can
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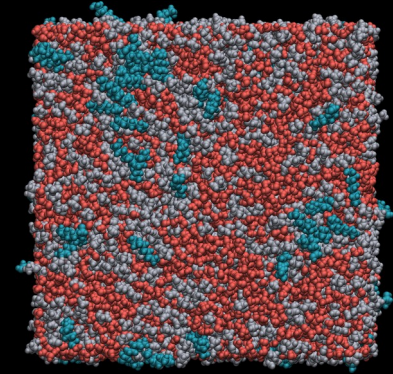
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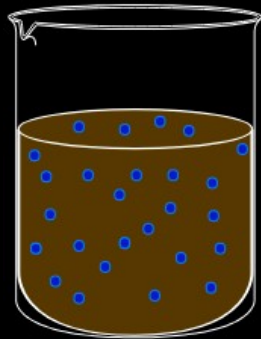
MD Simulation



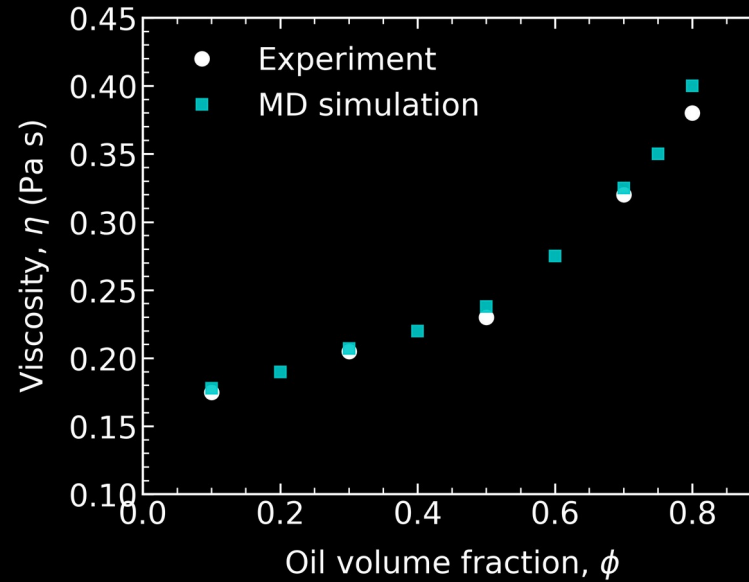
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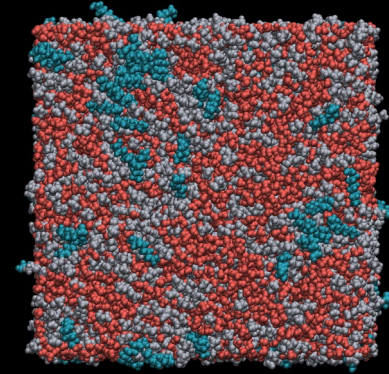
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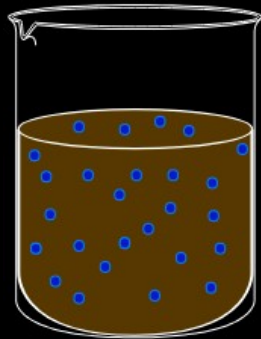
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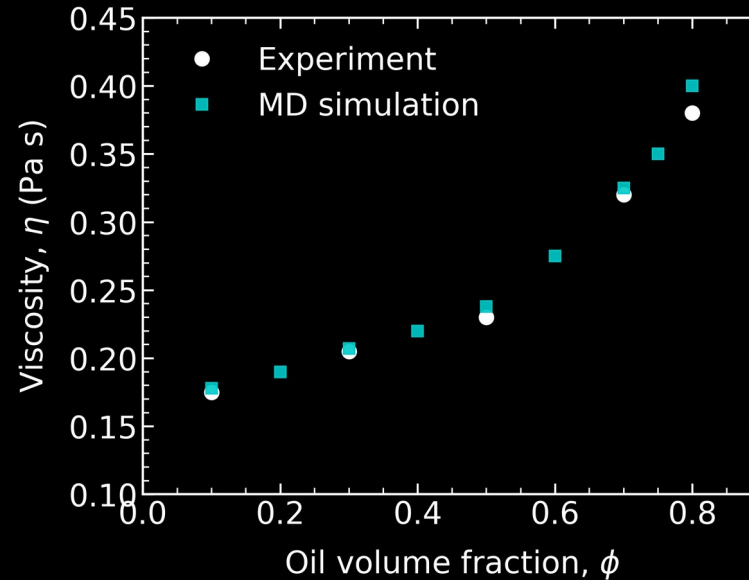
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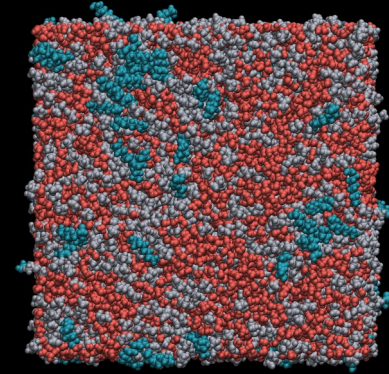


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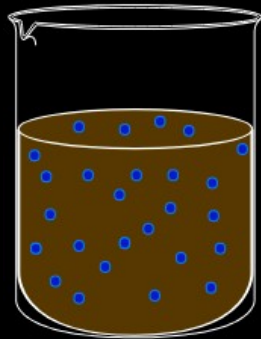
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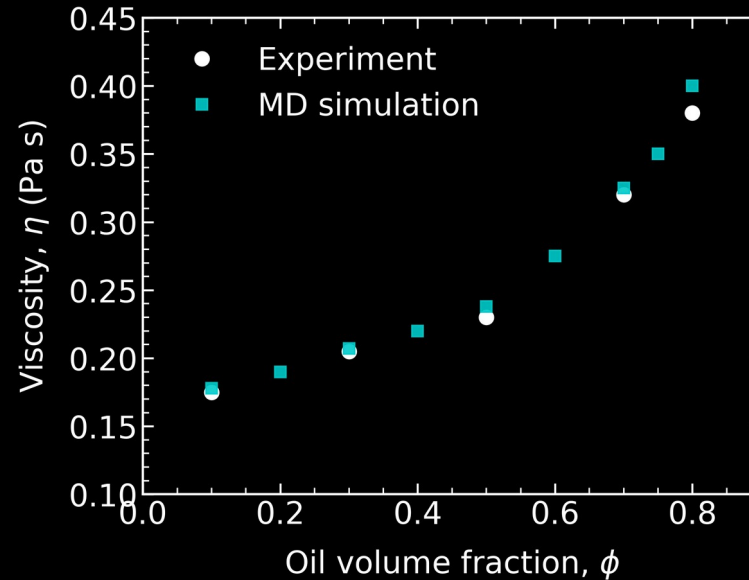
MD simulations
provide trajectories

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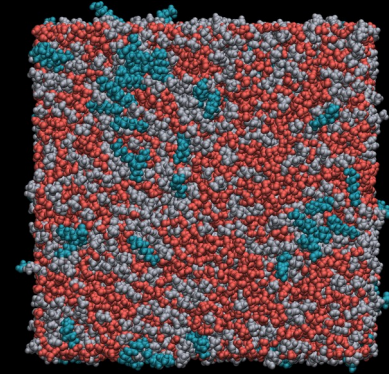
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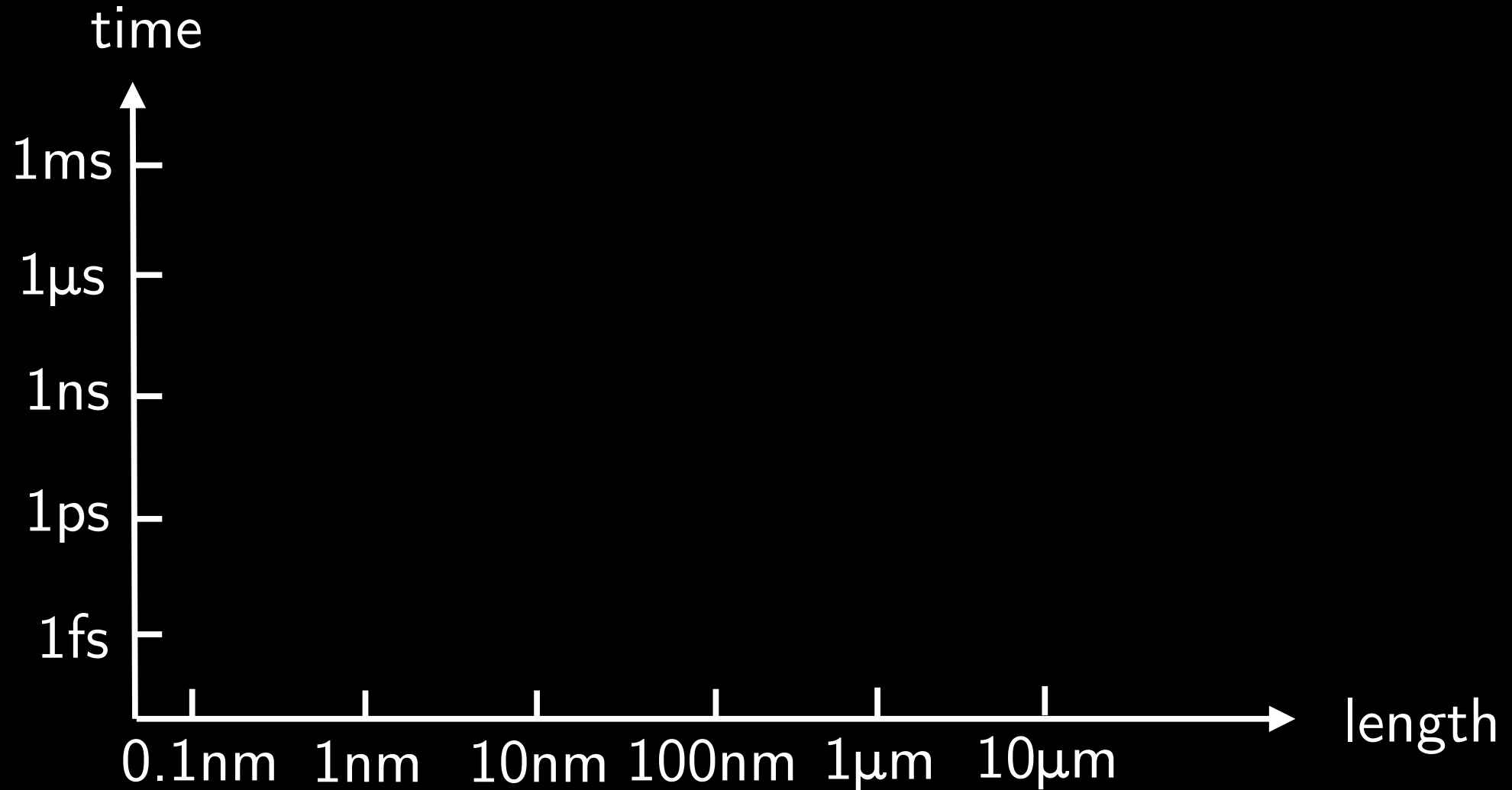


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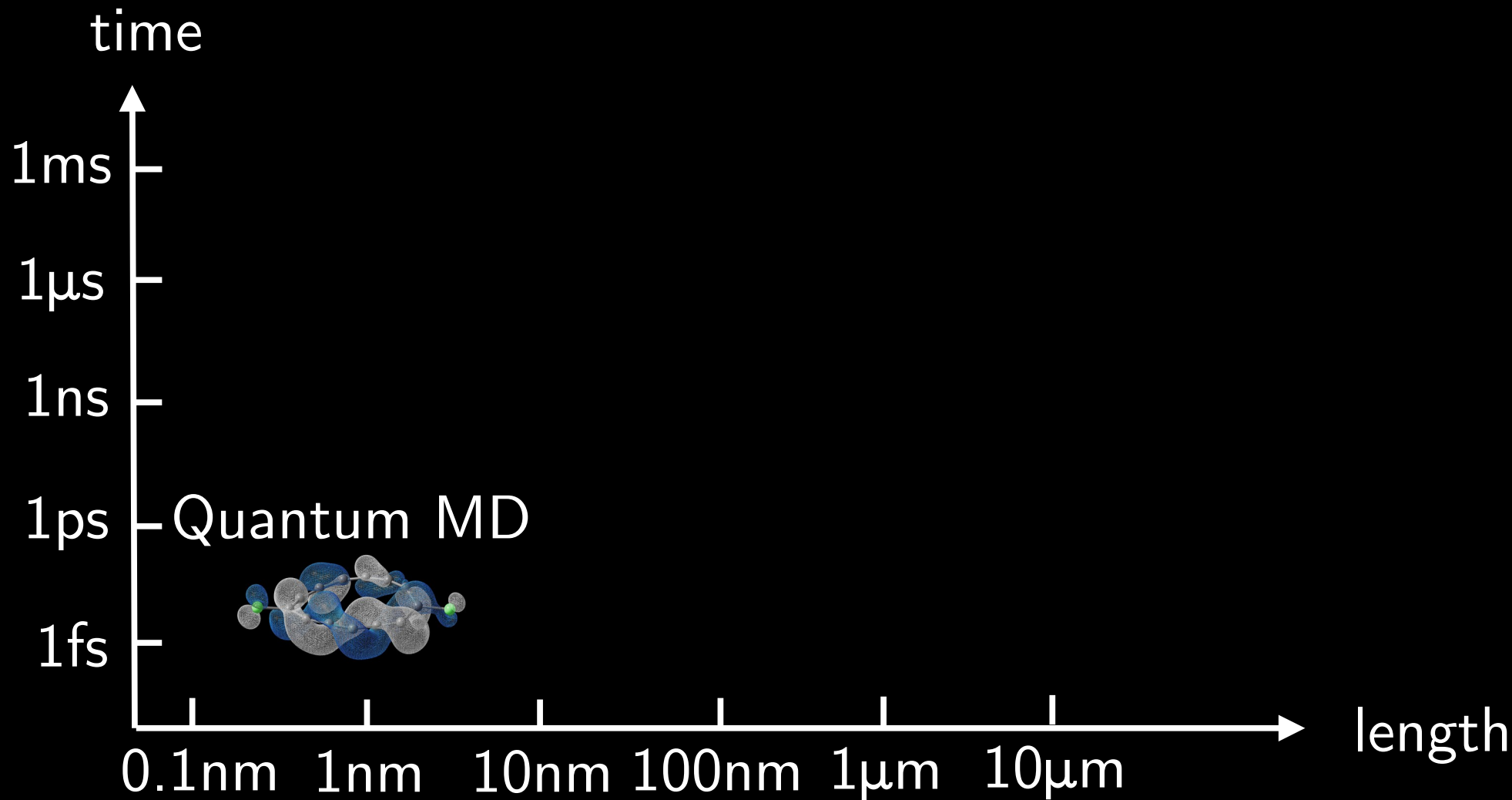
MD simulations
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Experiments are necessary to validate simulations.
Simulations can help interpret the results.

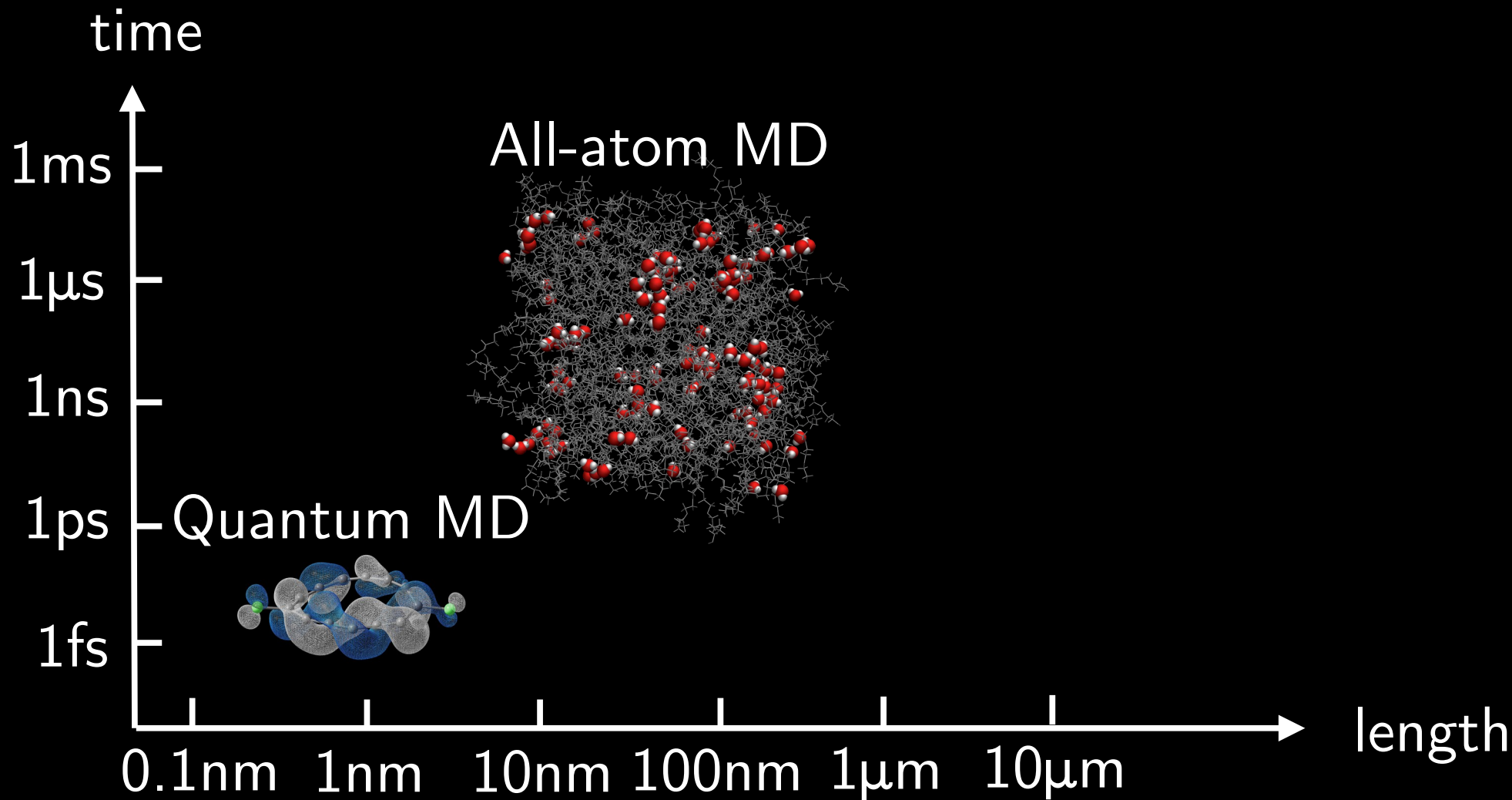
Time and length scales



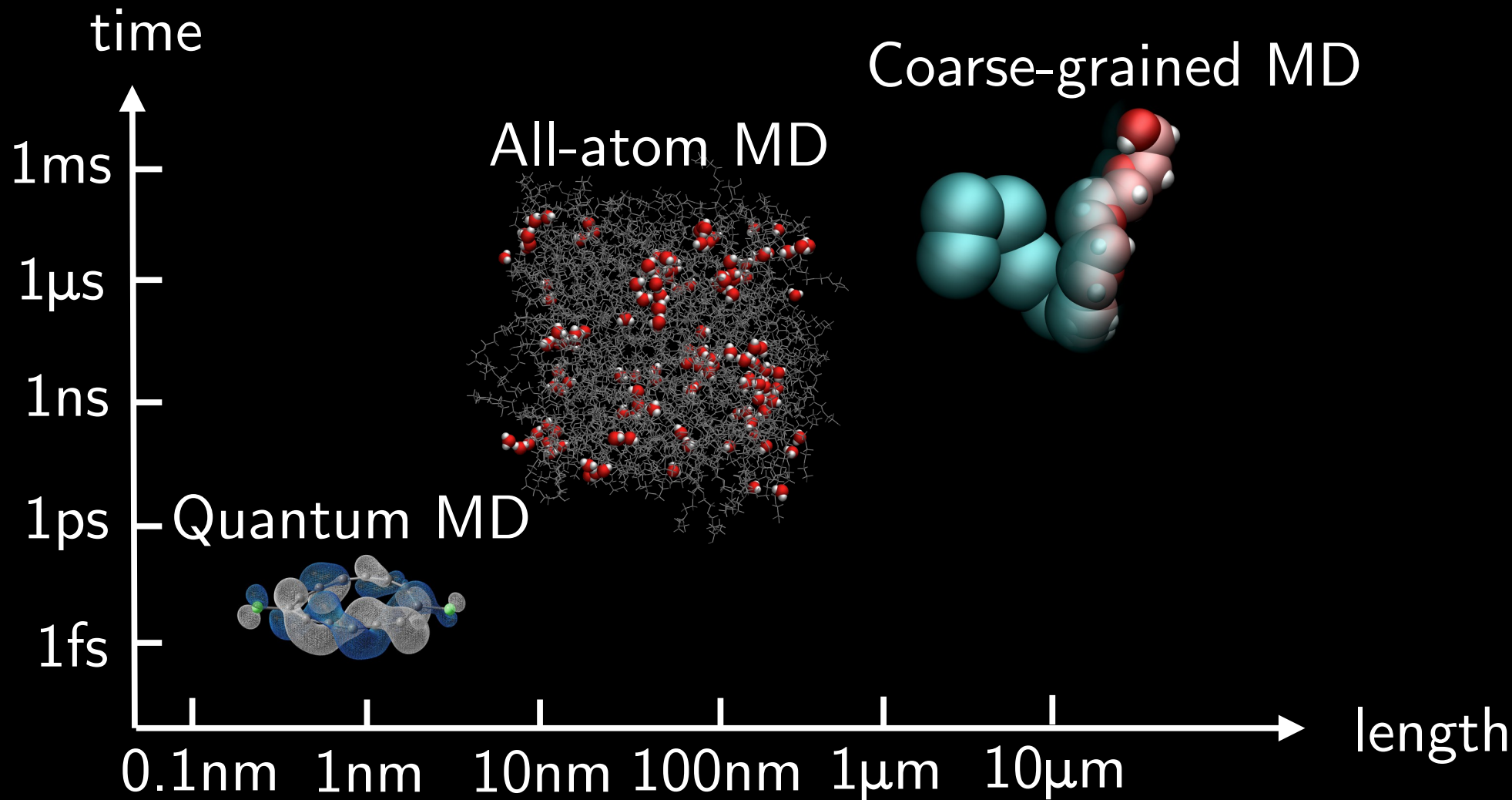
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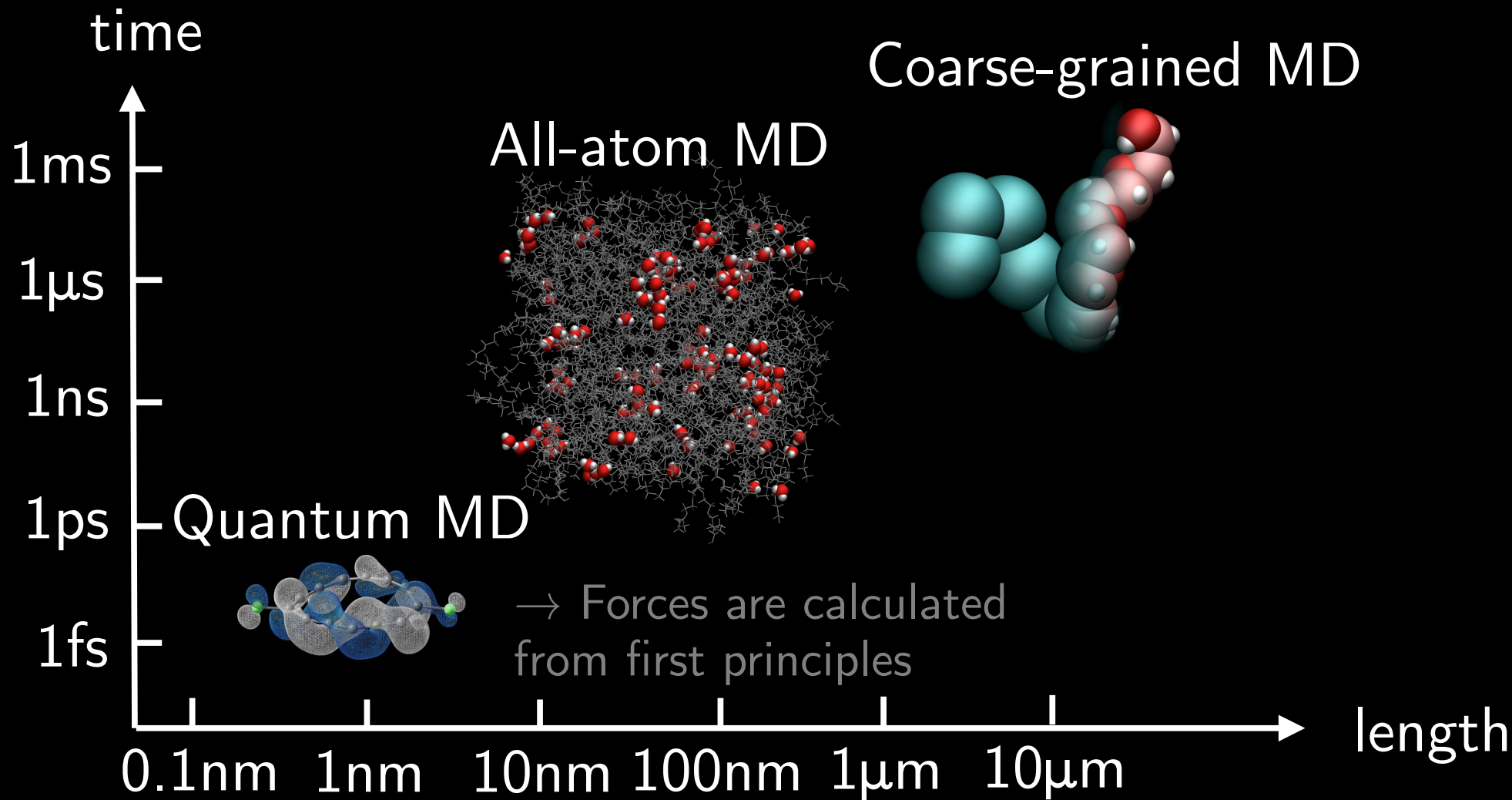
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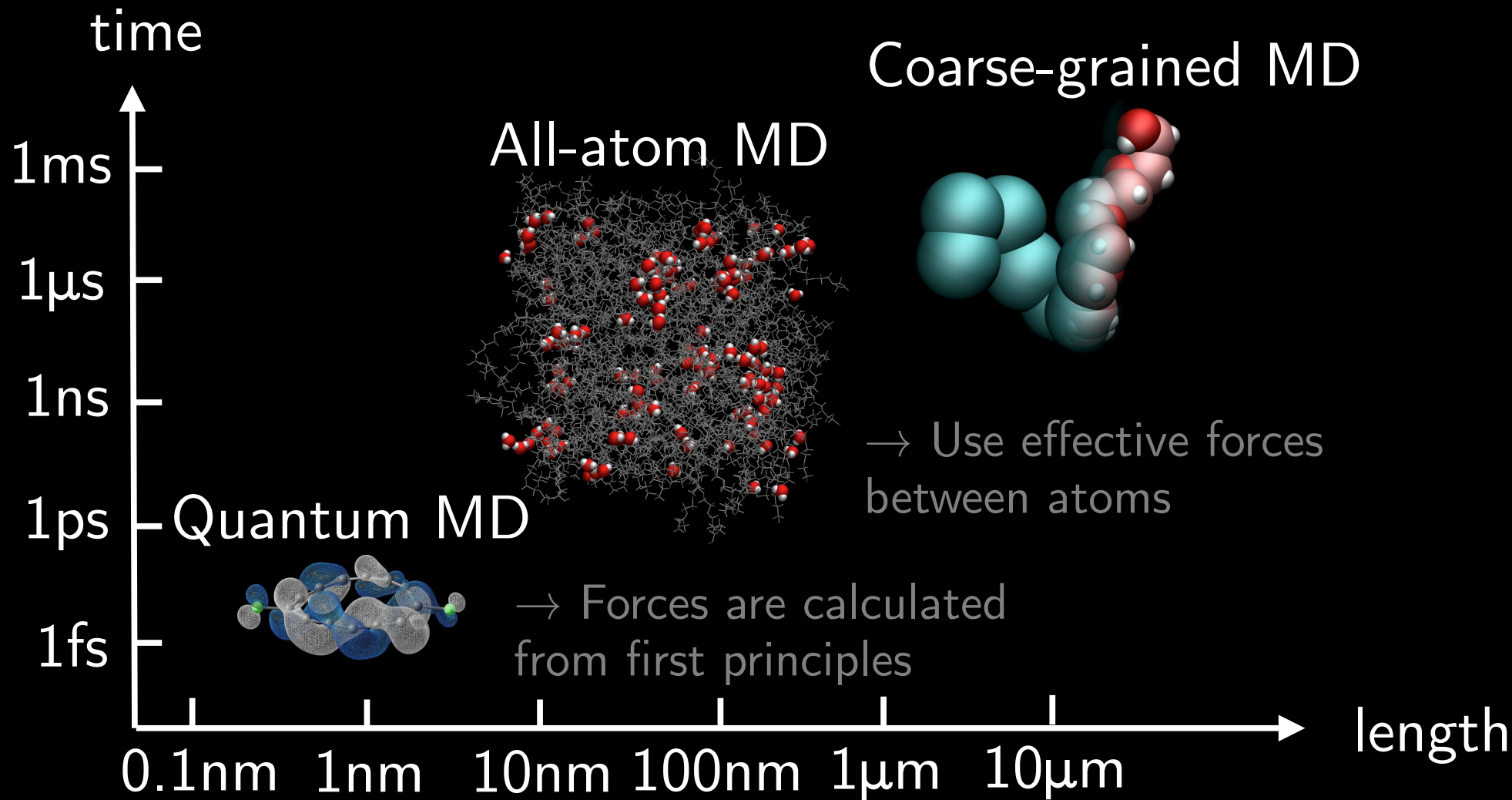
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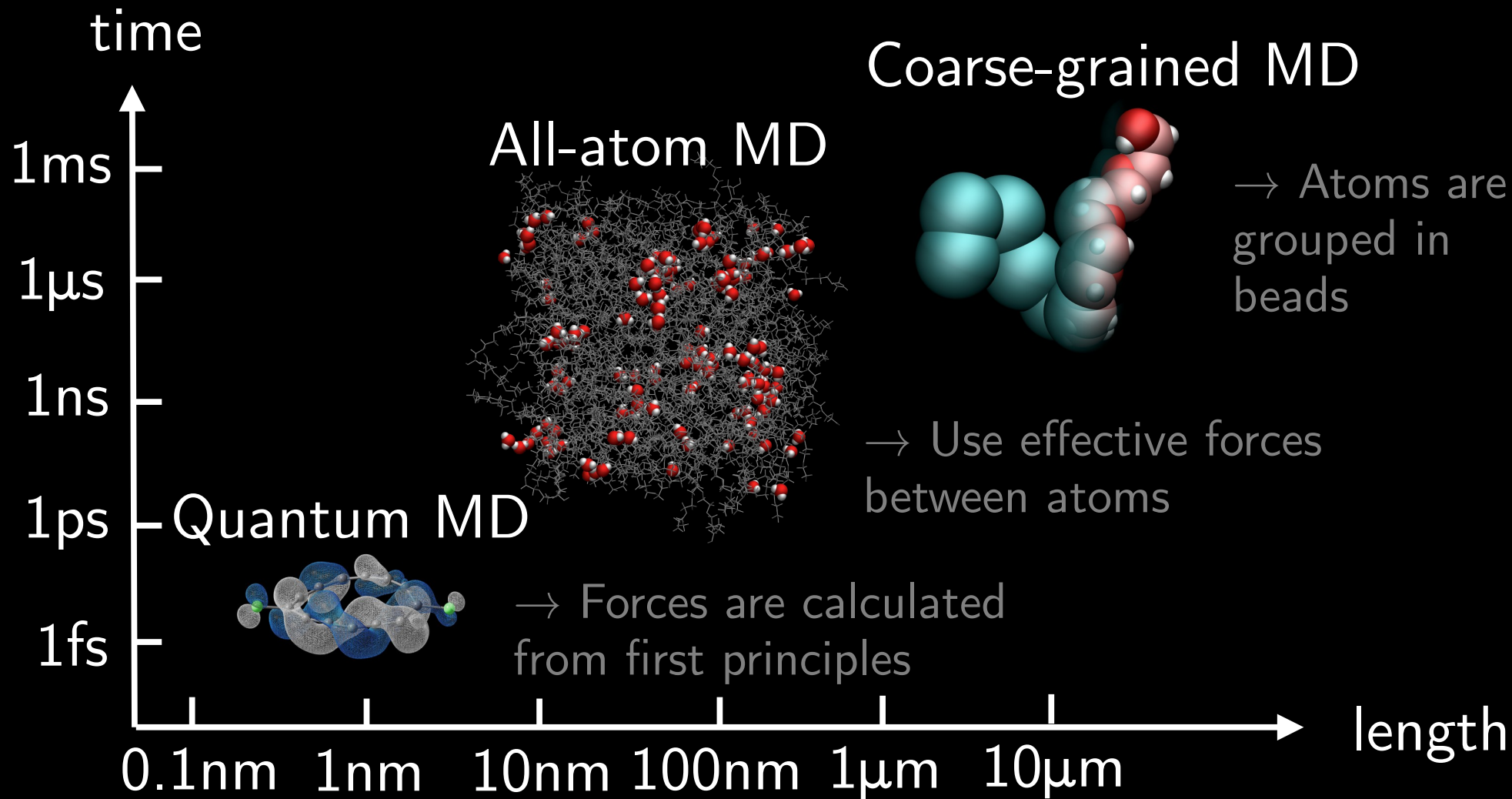
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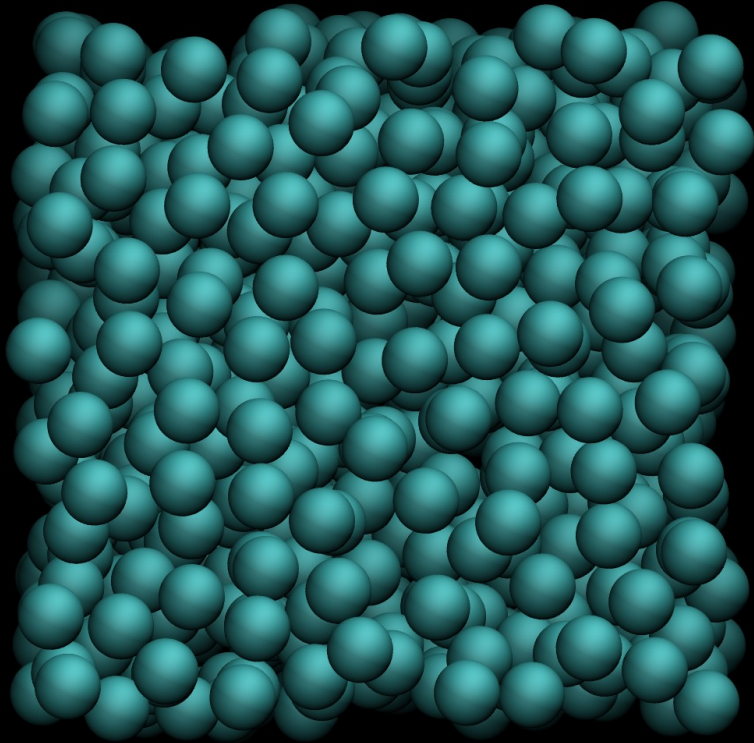


Time and length scales



Molecular dynamics

1)



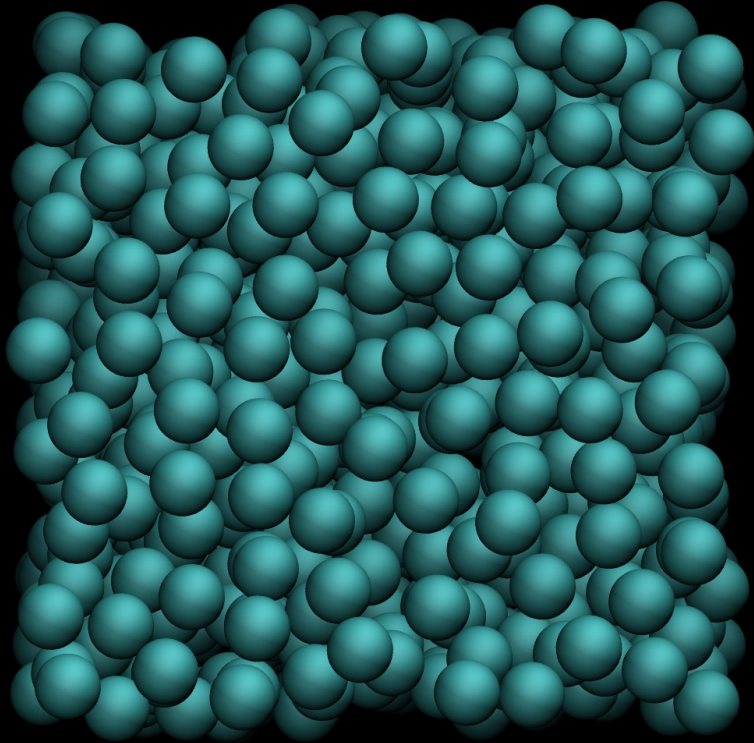
2)

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Molecular dynamics

1) Define the system



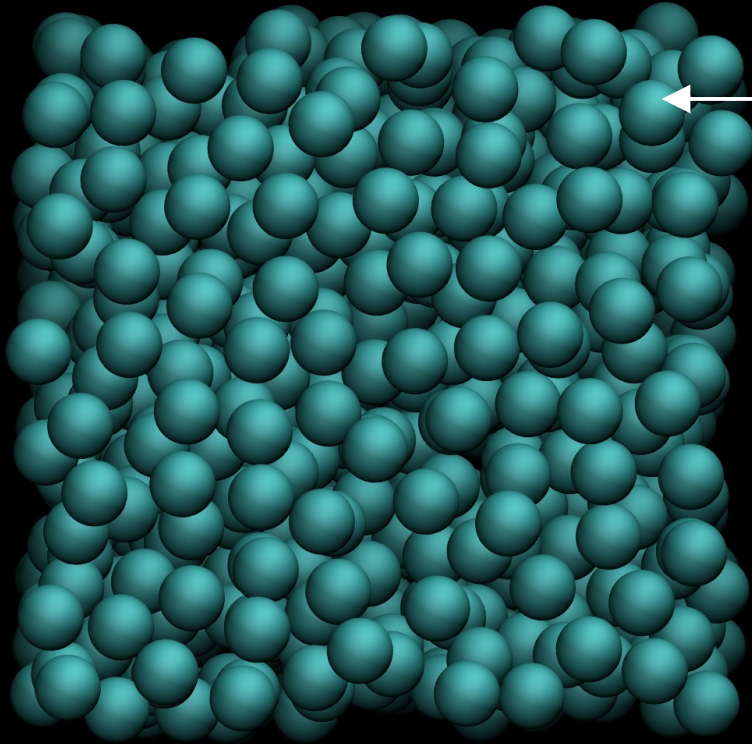
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Atom i (position $\mathbf{r}_i$, velocity $\dot{\mathbf{r}}_i$)

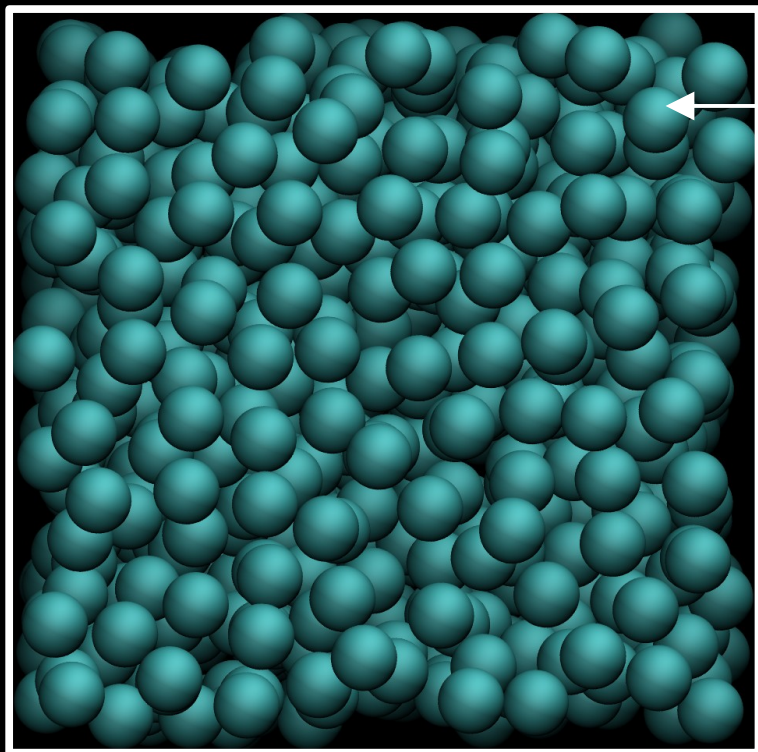
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Atom i (position $\mathbf{r}_i$, velocity $\dot{\mathbf{r}}_i$)

Box boundary (generally periodic)

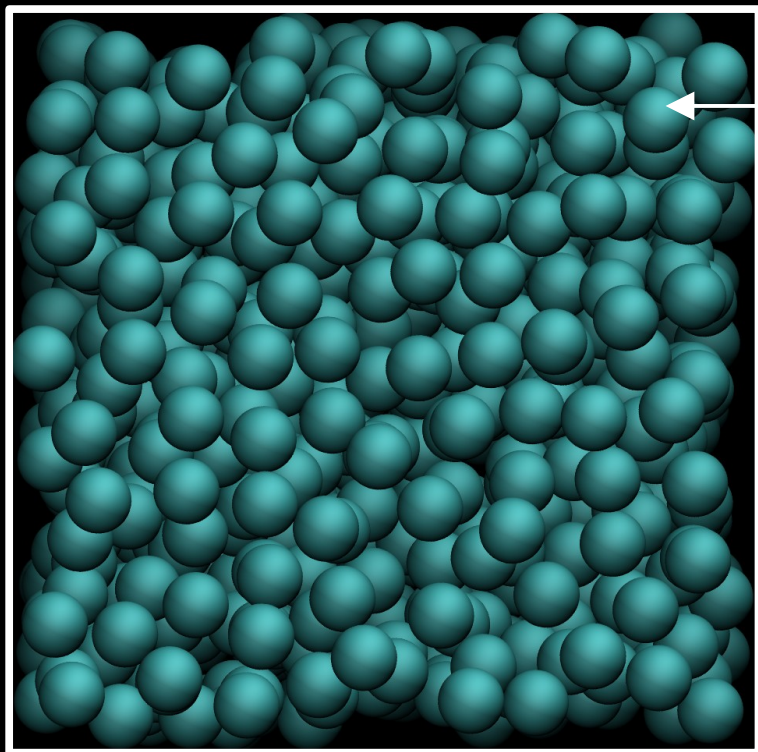
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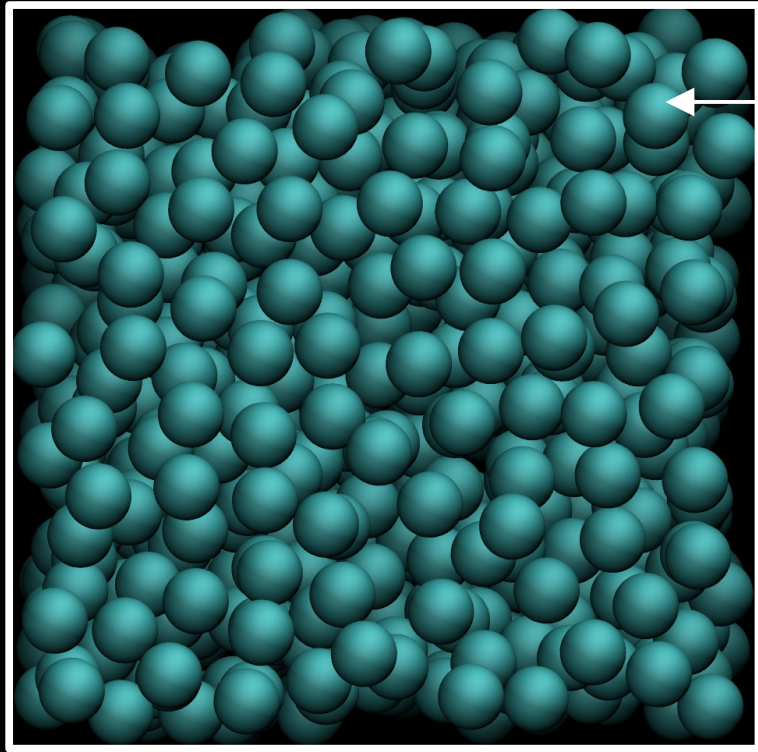
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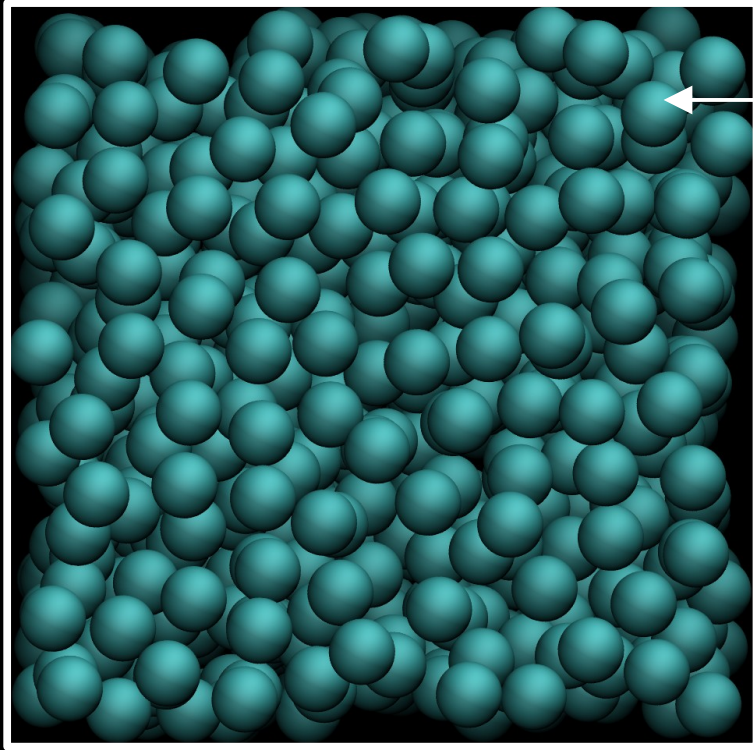
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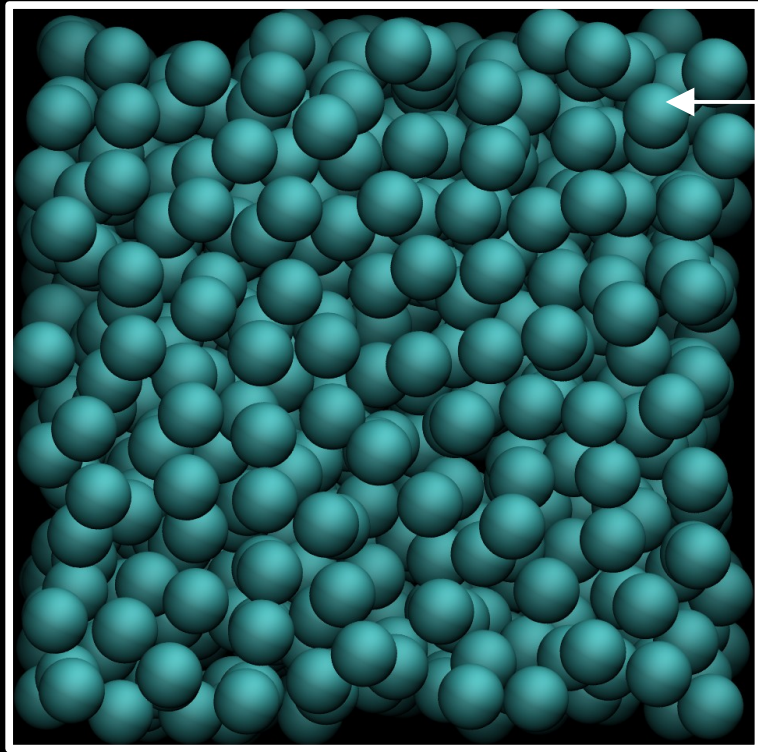
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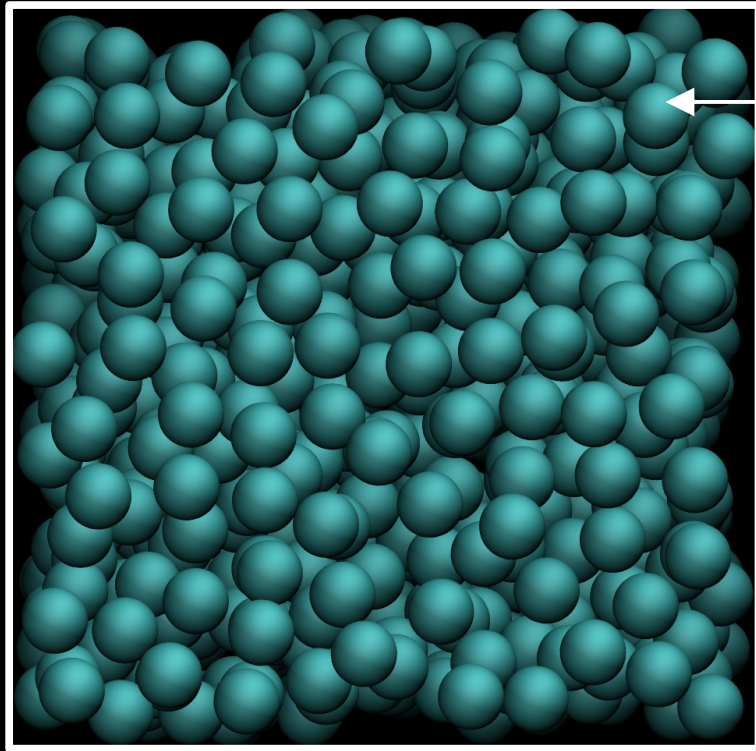
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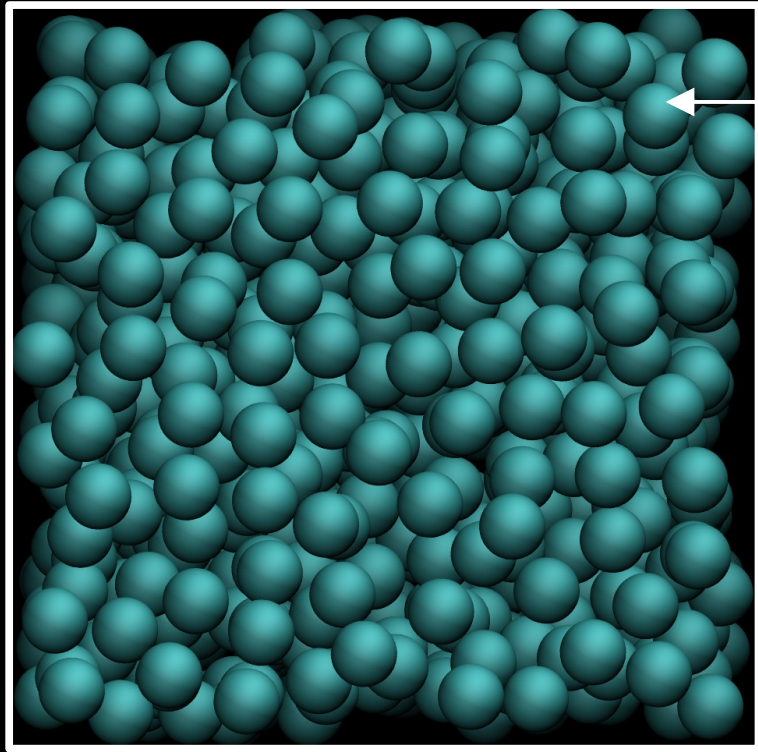
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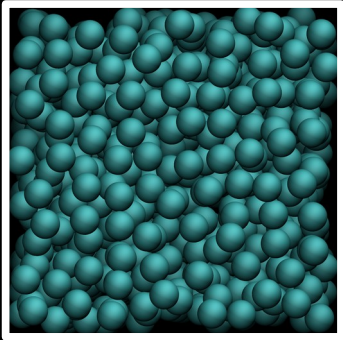
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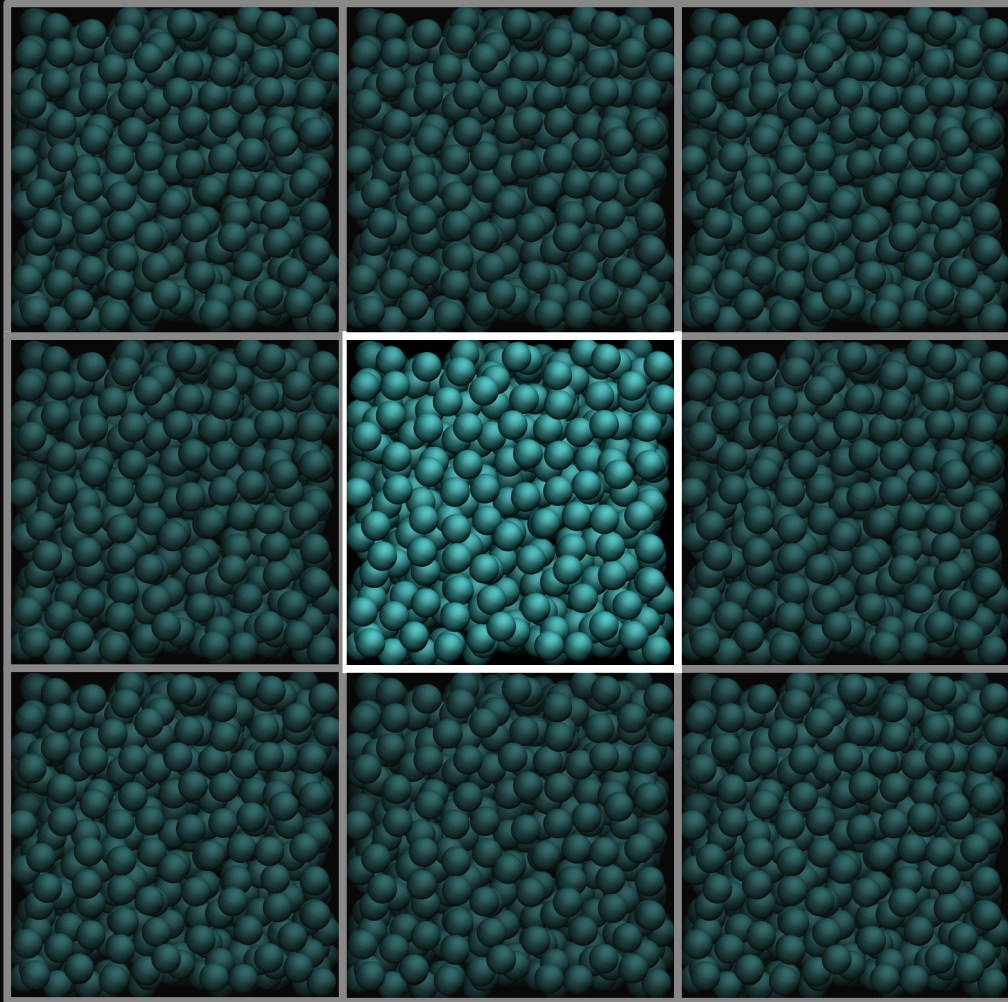
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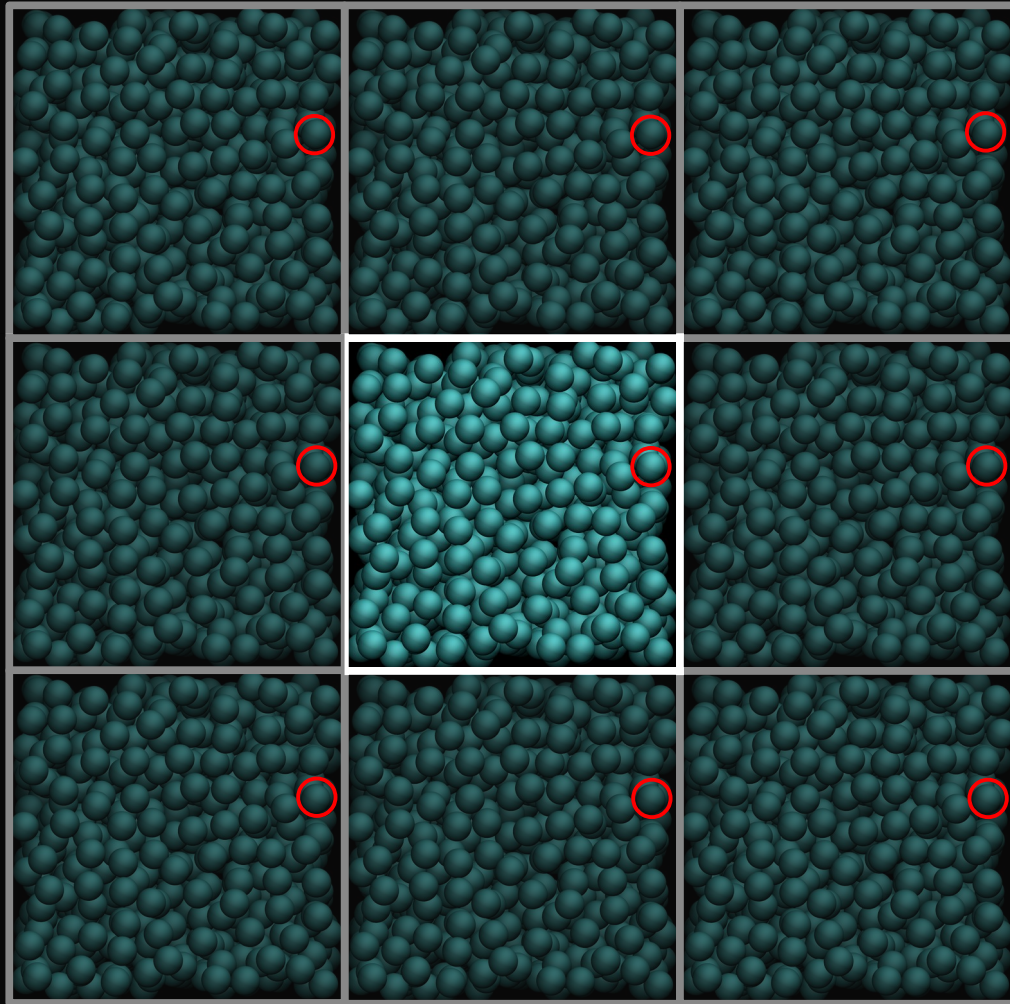
Periodic boundary conditions



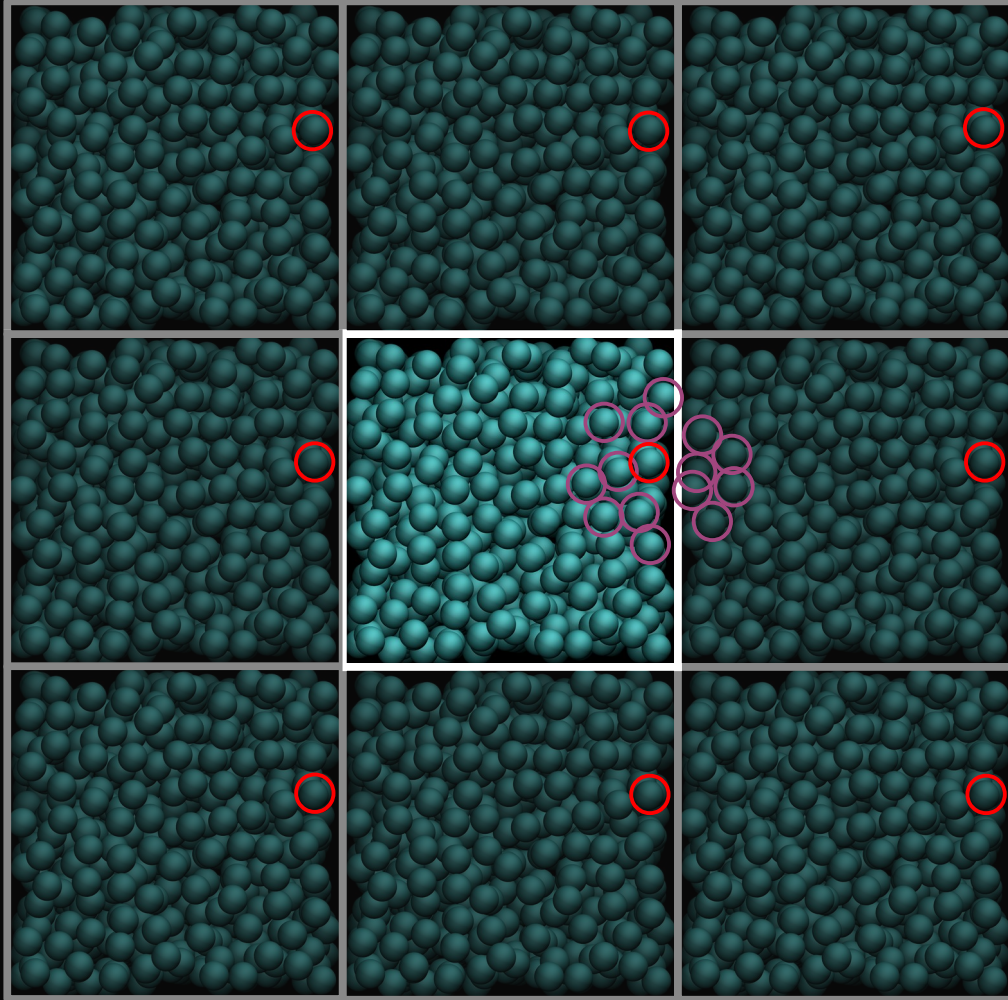
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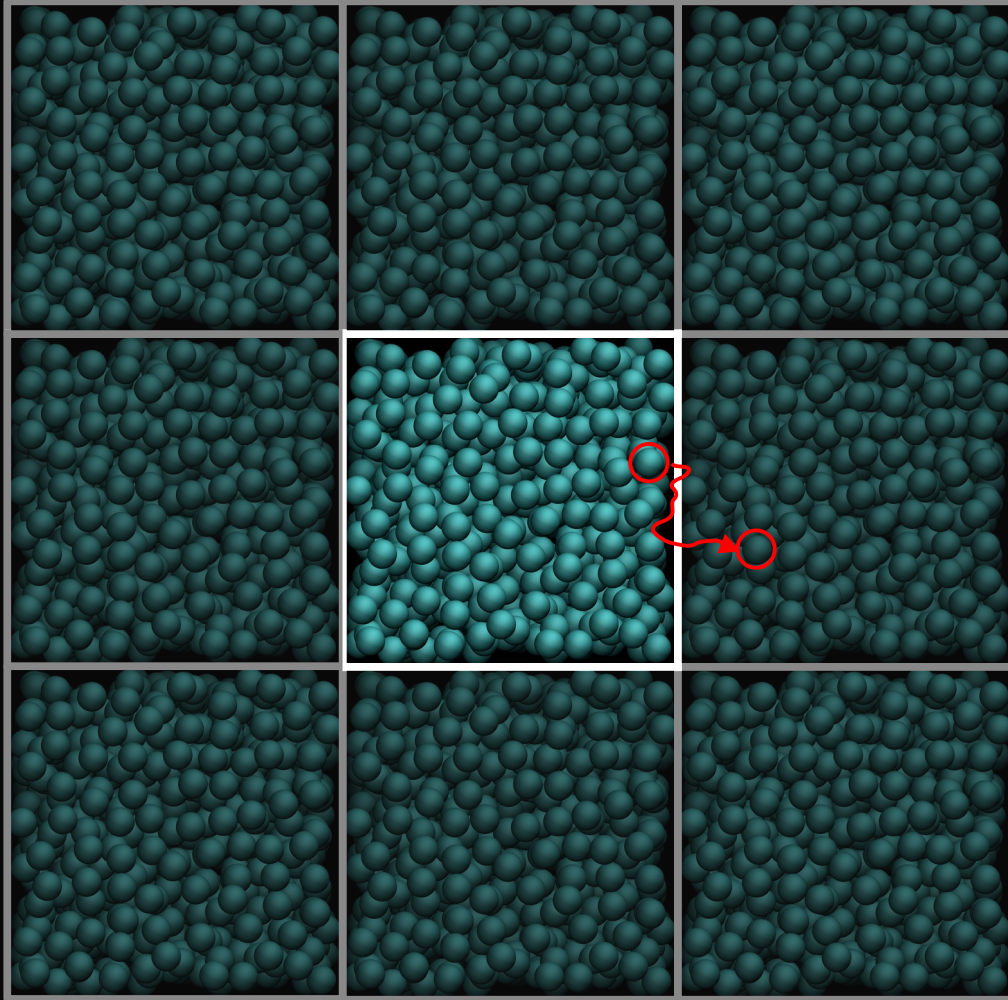
Periodic boundary conditions



Atoms:

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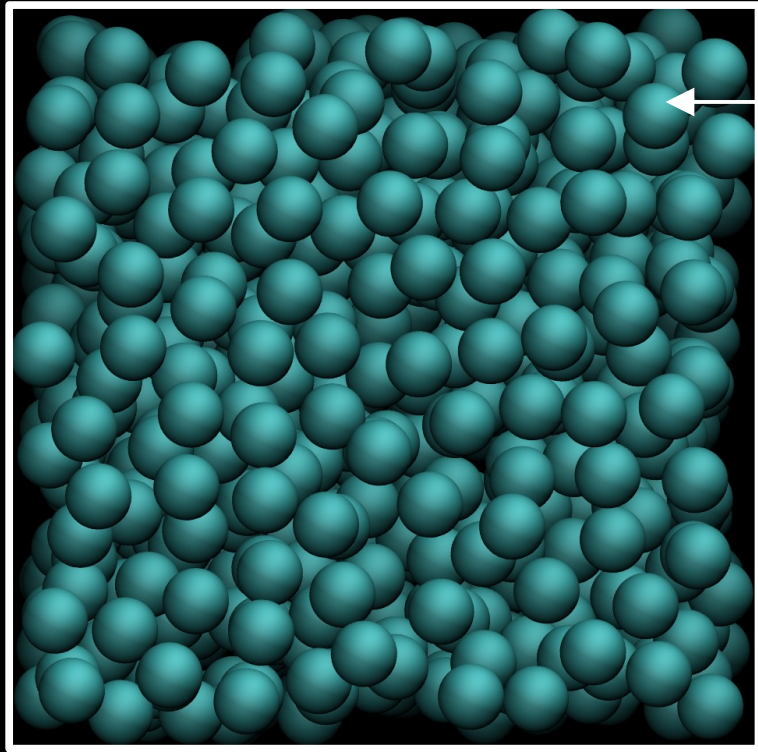
Periodic boundary conditions



Atoms:

- interact through the boundaries
- can cross the boundaries

Molecular dynamics



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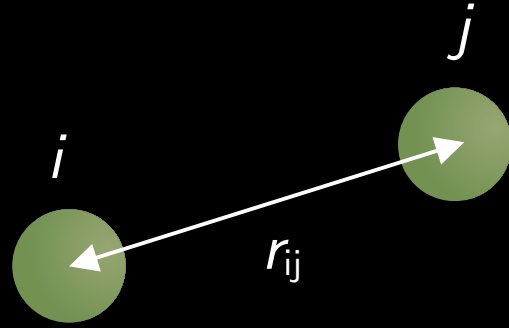
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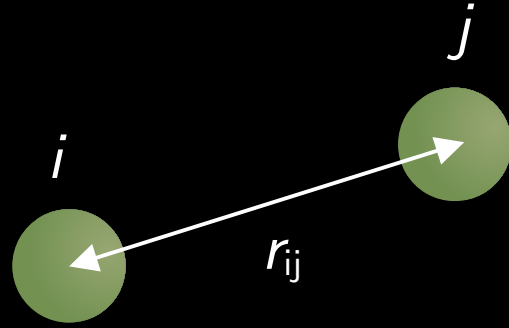
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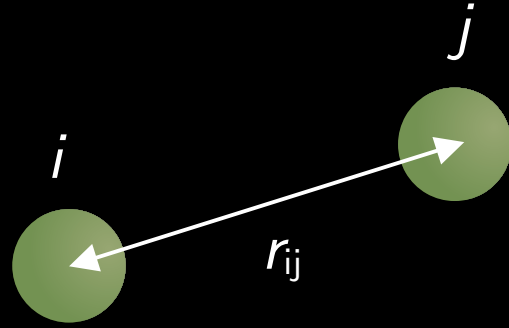


Forces between atoms

Conventional potential : Lennard Jones

$$F_{ij} = -\frac{dU(r_{ij})}{dr_{ij}}$$

$$U_{ij}(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$

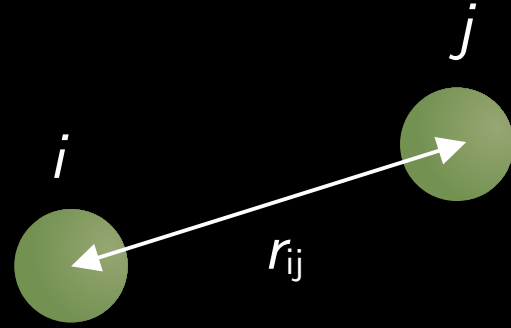


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σ_{ij} : effective particle size

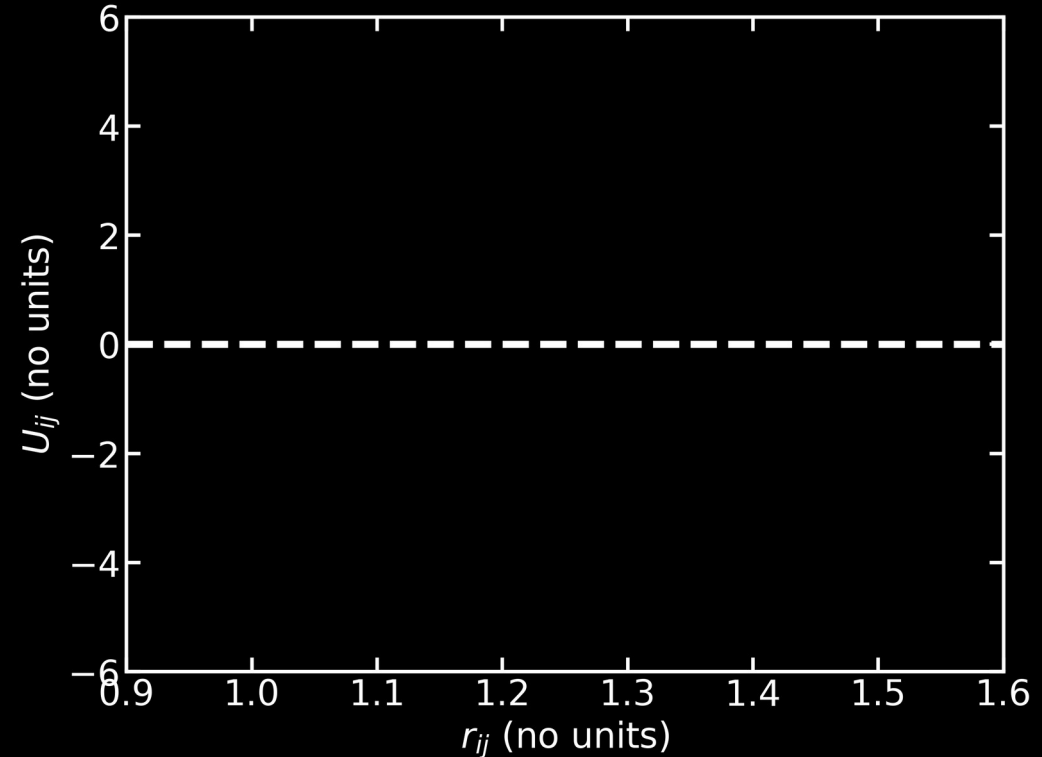
ϵ_{ij} : interaction strength

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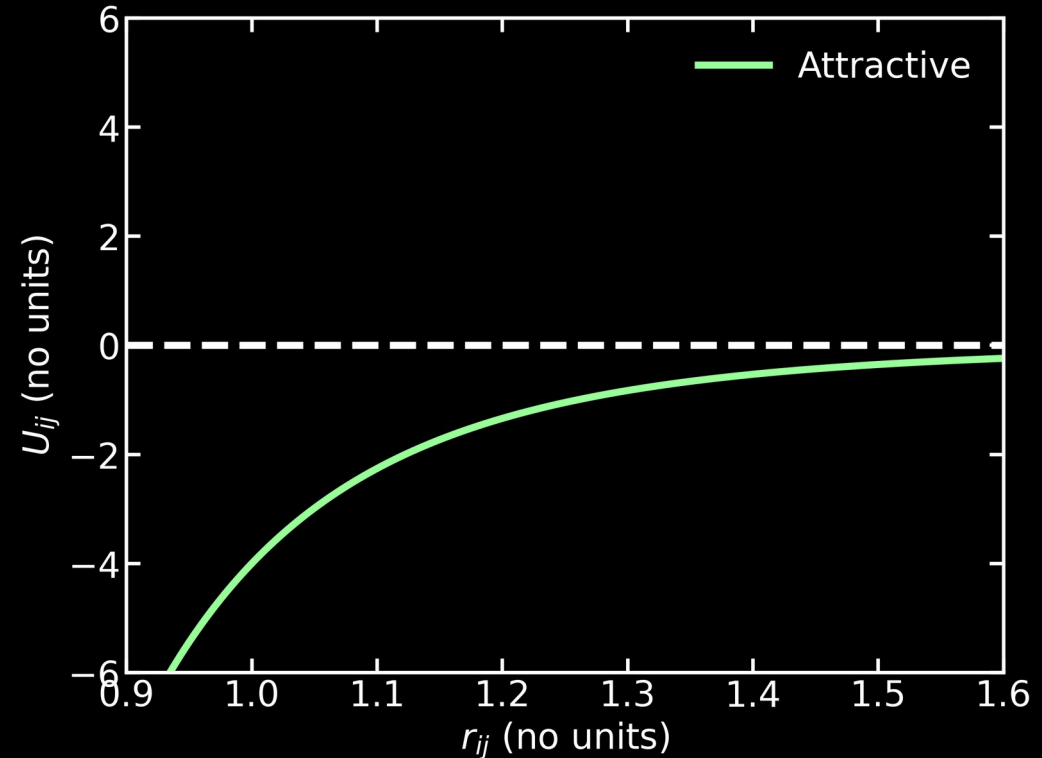
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Attractive (London dispersion)



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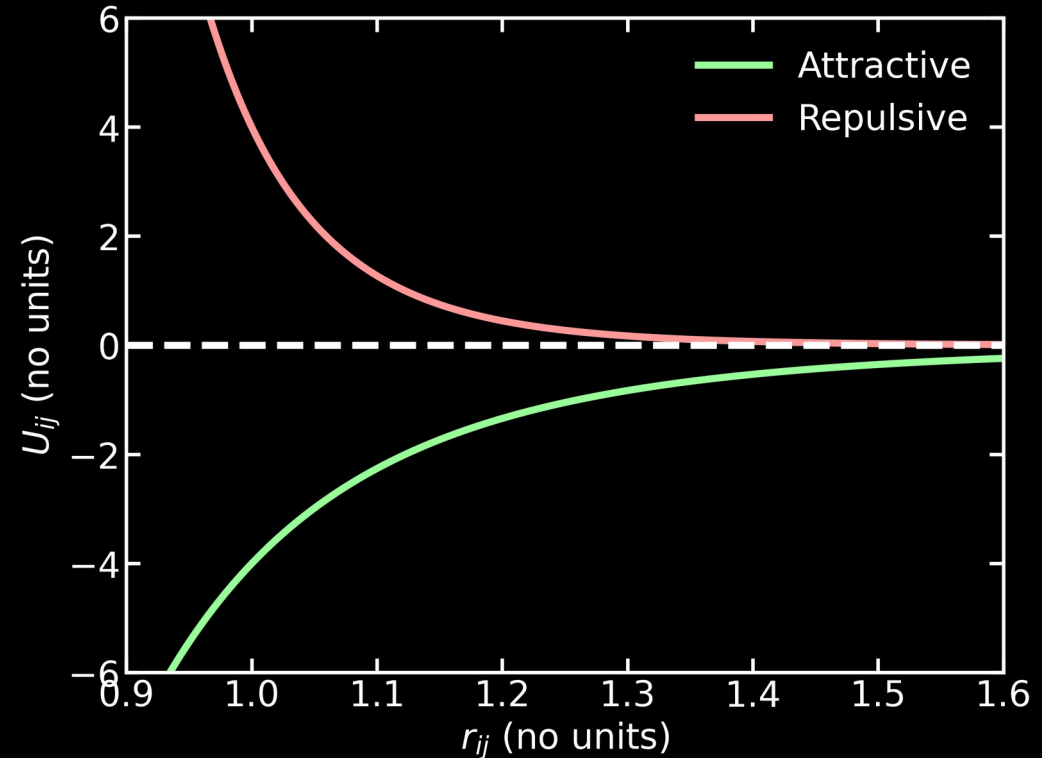
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Repulsive (Pauli repulsion)

Attractive (London dispersion)



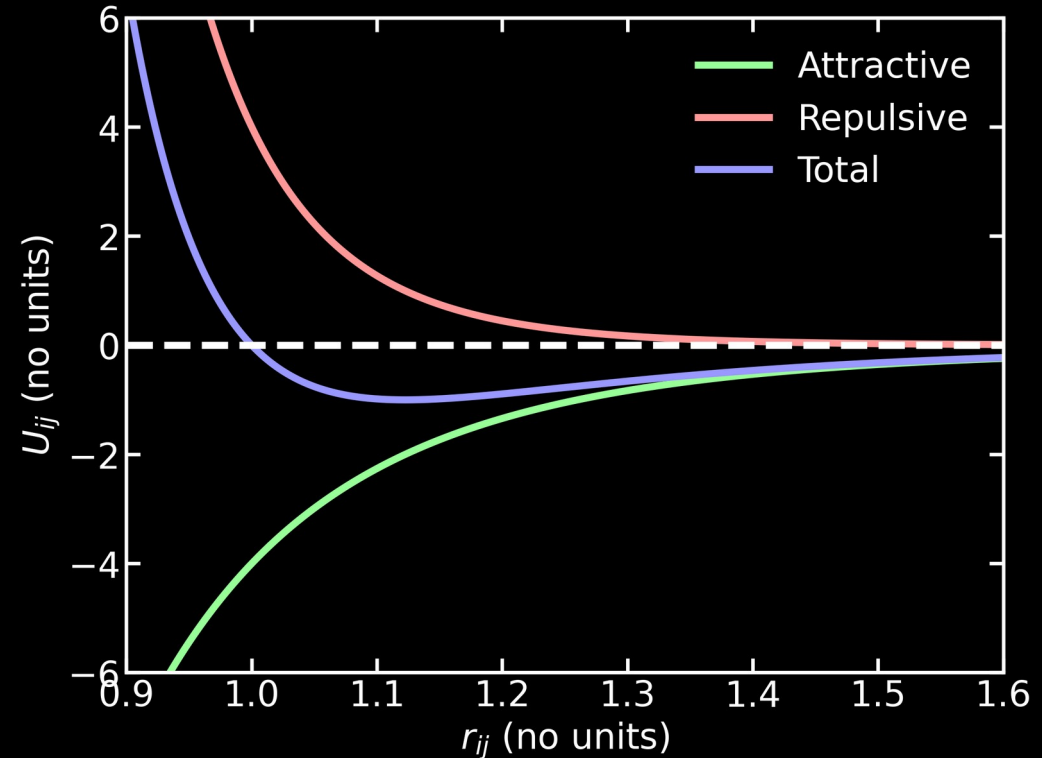
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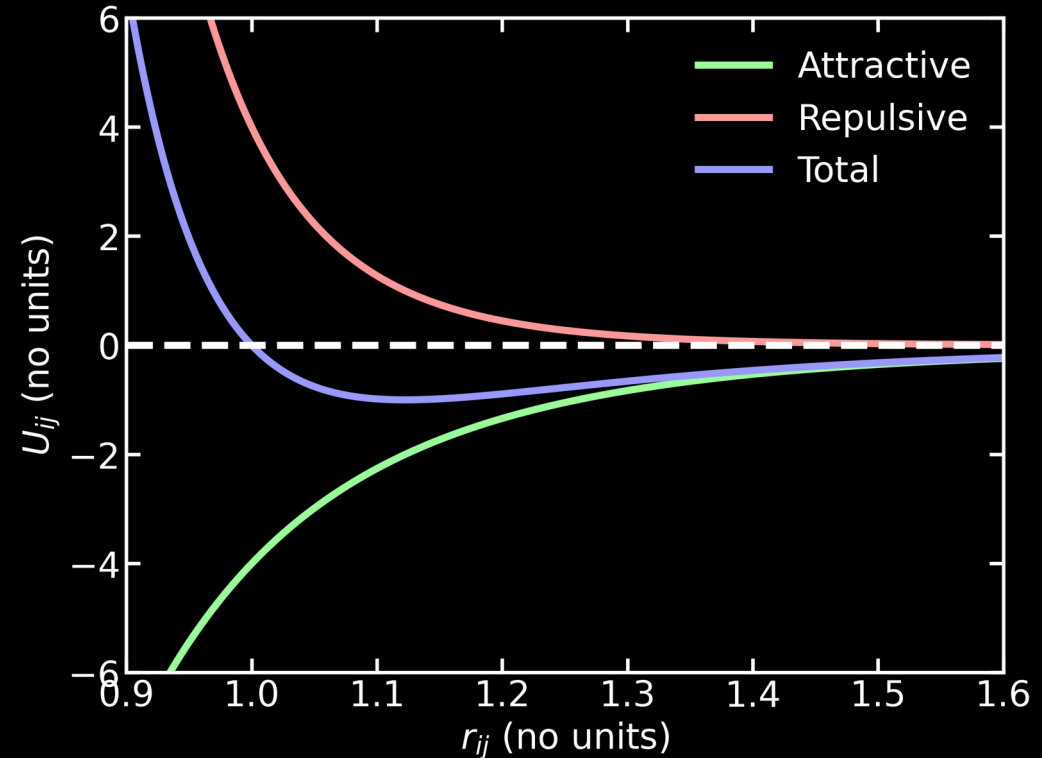
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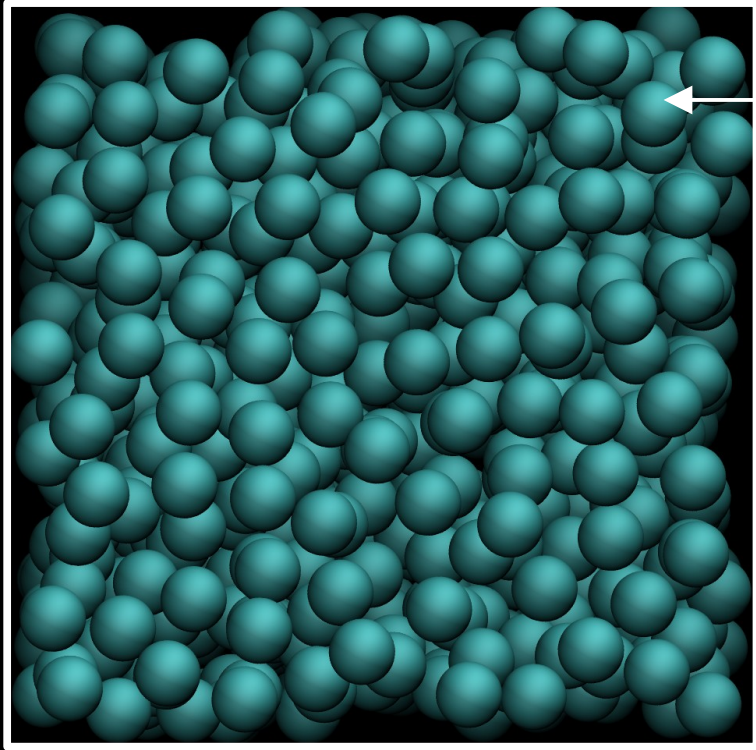
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Generally only calculated for atoms within a certain distance



Molecular dynamics



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Box boundary (generally periodic)

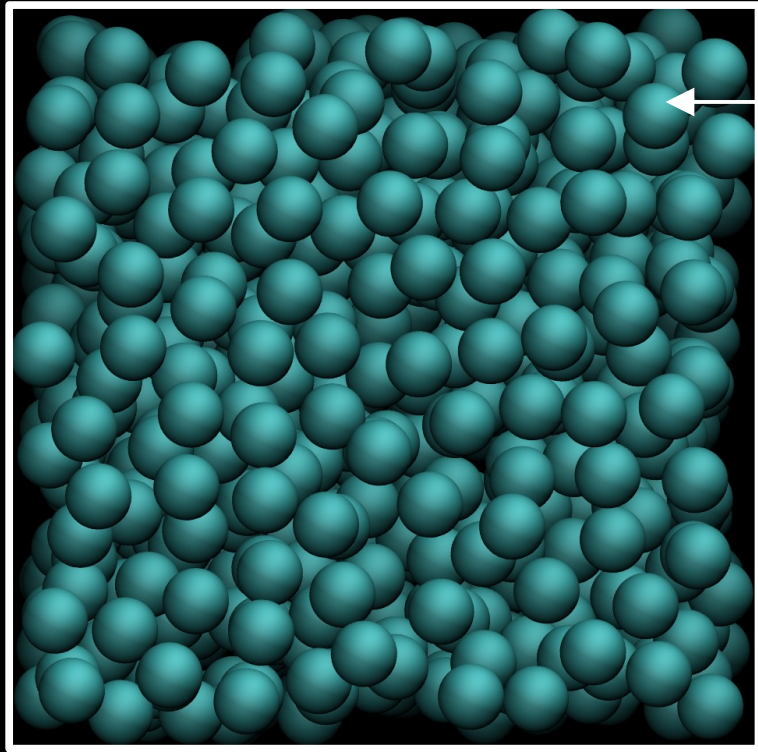
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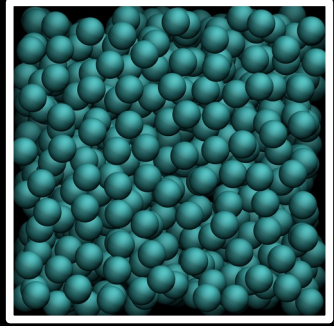
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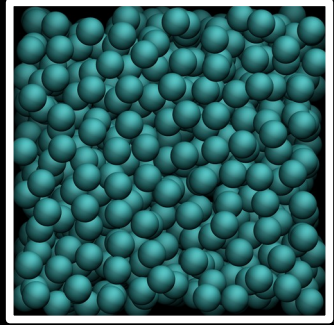
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Solving Newton's equation



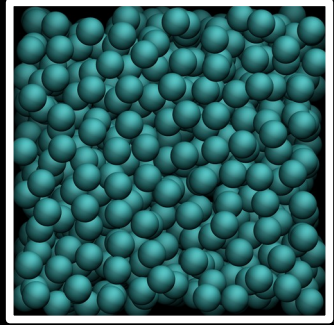
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Solving Newton's equation



$$\sum_{j \neq i} \mathbf{F}_{ij} = m_i \ddot{\mathbf{r}}_i \rightarrow \text{Continuous differential equation}$$

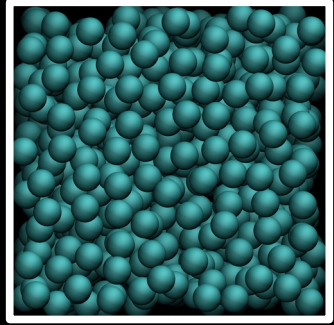
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Verlet algorithm : numerical way to solve Newton's equation

Solving Newton's equation

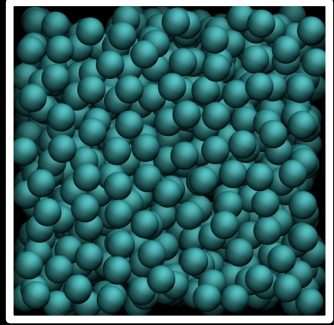


$$\sum_{j \neq i} \mathbf{F}_{ij} = m_i \ddot{\mathbf{r}}_i \rightarrow \text{Continuous differential equation}$$

Verlet algorithm : numerical way to solve Newton's equation

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

Solving Newton's equation



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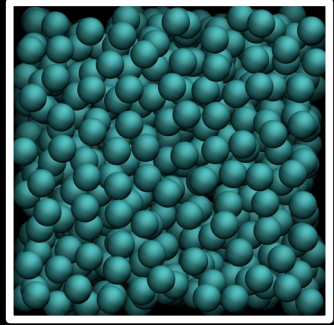
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Timestep



Solving Newton's equation



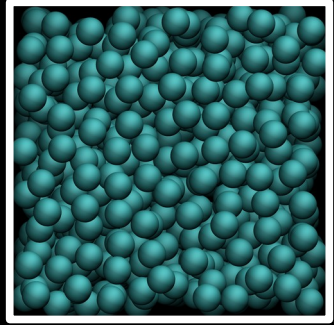
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$$\dot{\mathbf{r}}(t + \Delta t) = \dot{\mathbf{r}}(t) + \frac{\ddot{\mathbf{r}}(t) + \ddot{\mathbf{r}}(t + \Delta t)}{2}\Delta t$$

Solving Newton's equation



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Verlet algorithm : numerical way to solve Newton's equation

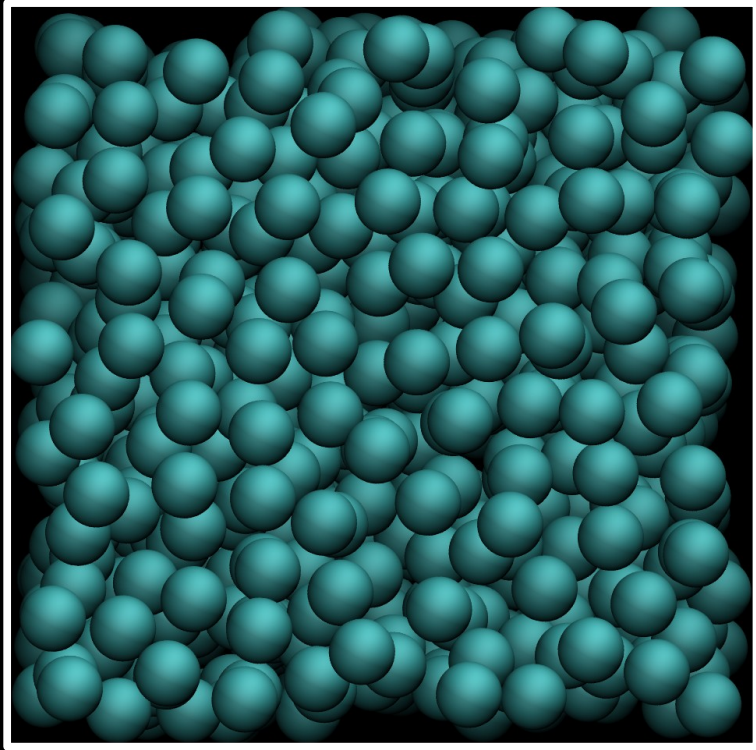
$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

$$\dot{\mathbf{r}}(t + \Delta t) = \dot{\mathbf{r}}(t) + \frac{\ddot{\mathbf{r}}(t) + \ddot{\mathbf{r}}(t + \Delta t)}{2}\Delta t$$

Taylor expansion of the positions at time $t + \Delta t$:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2!}\ddot{\mathbf{r}}(t)\Delta t^2 + \frac{1}{3!}\ddot{\mathbf{r}}^{(3)}(t)\Delta t^3 + \mathcal{O}(\Delta t^4)$$

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1) Define the system

2) Measure forces between atoms $\mathbf{F}_{ij}$

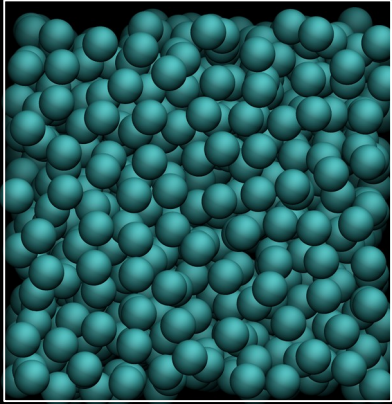
$$F_{ij} = -\frac{dU(r_{ij})}{dr_{ij}} \quad U_{ij}(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$

3) Solve Newton's equations of motion

4) Move atoms

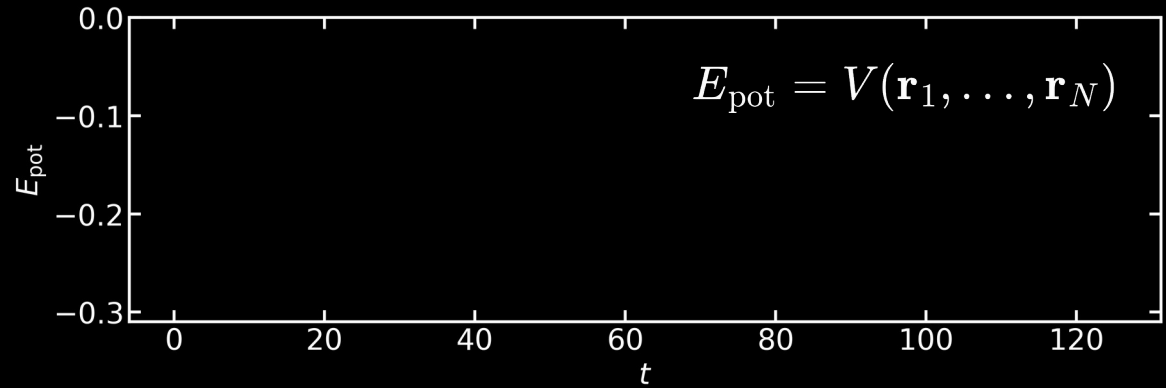
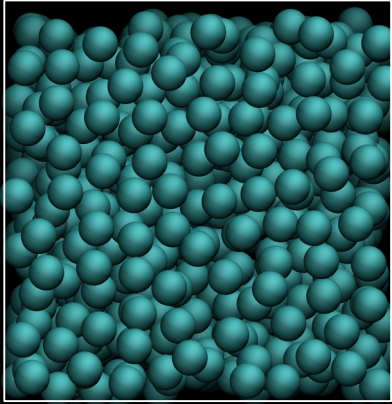
$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

Energies from MD



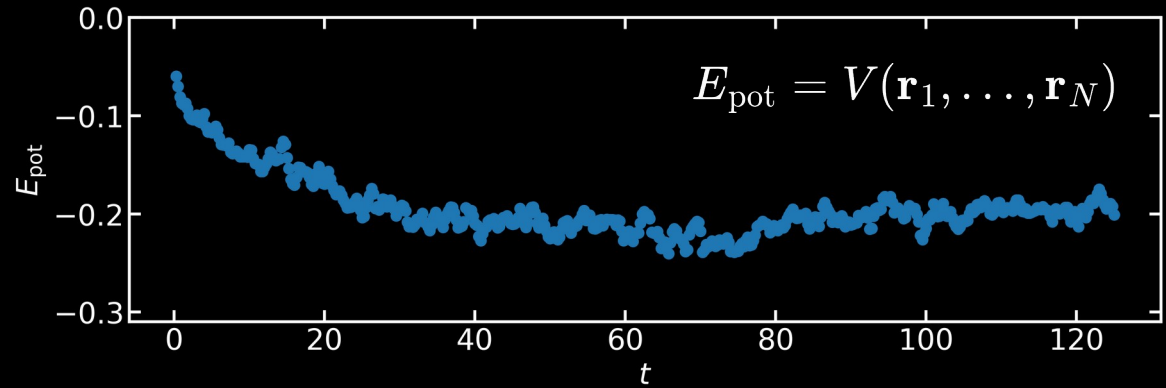
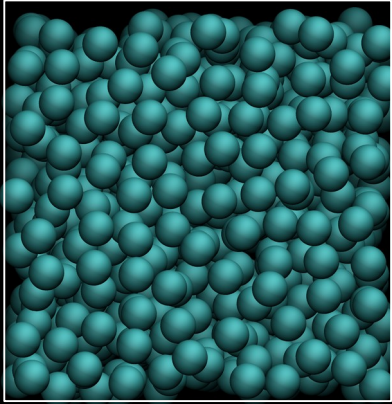
- 40 x 40 x 40 box
- 600 atoms
- Define LJ potential
- Apply Verlet algorithm

Energies from MD



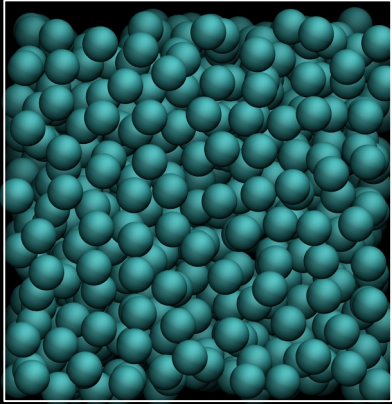
- 40 x 40 x 40 box
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Energies from MD

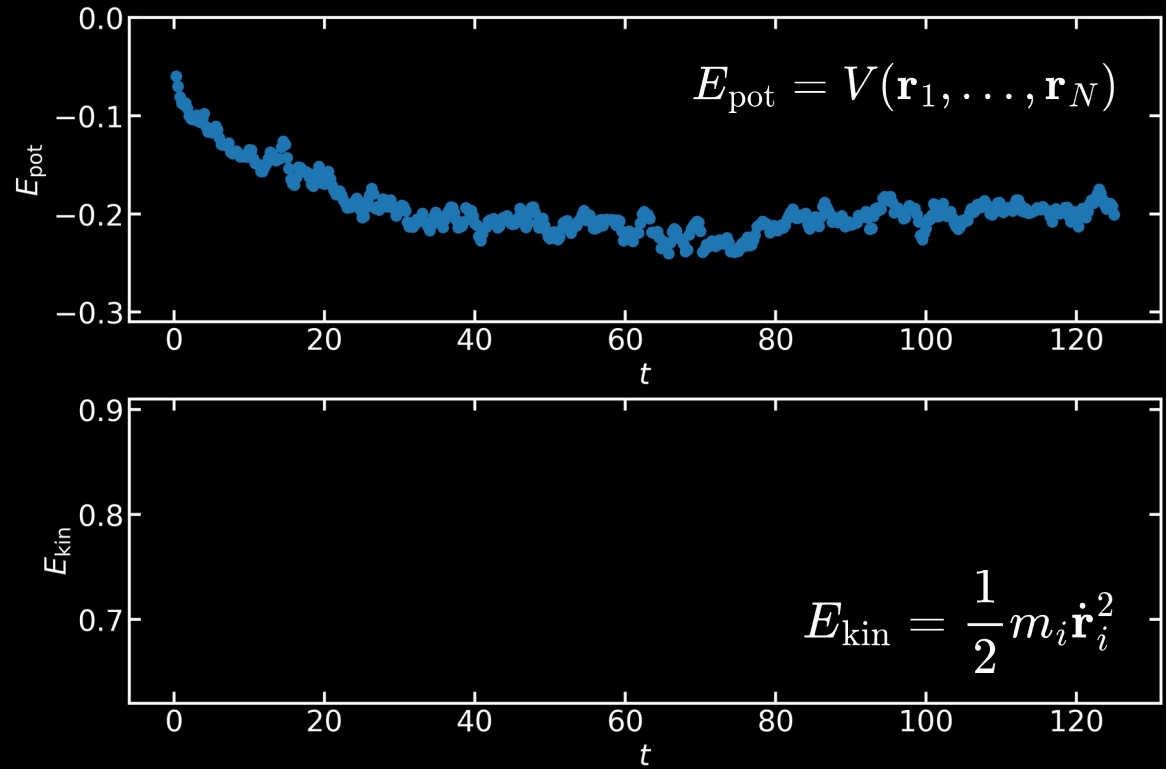


- 40 x 40 x 40 box
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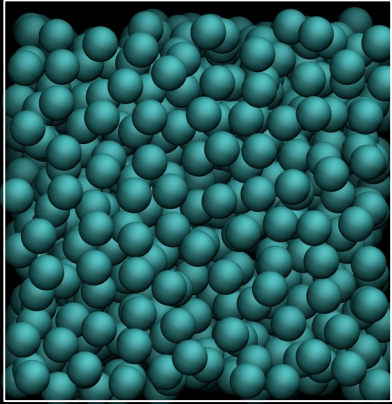
Energies from MD



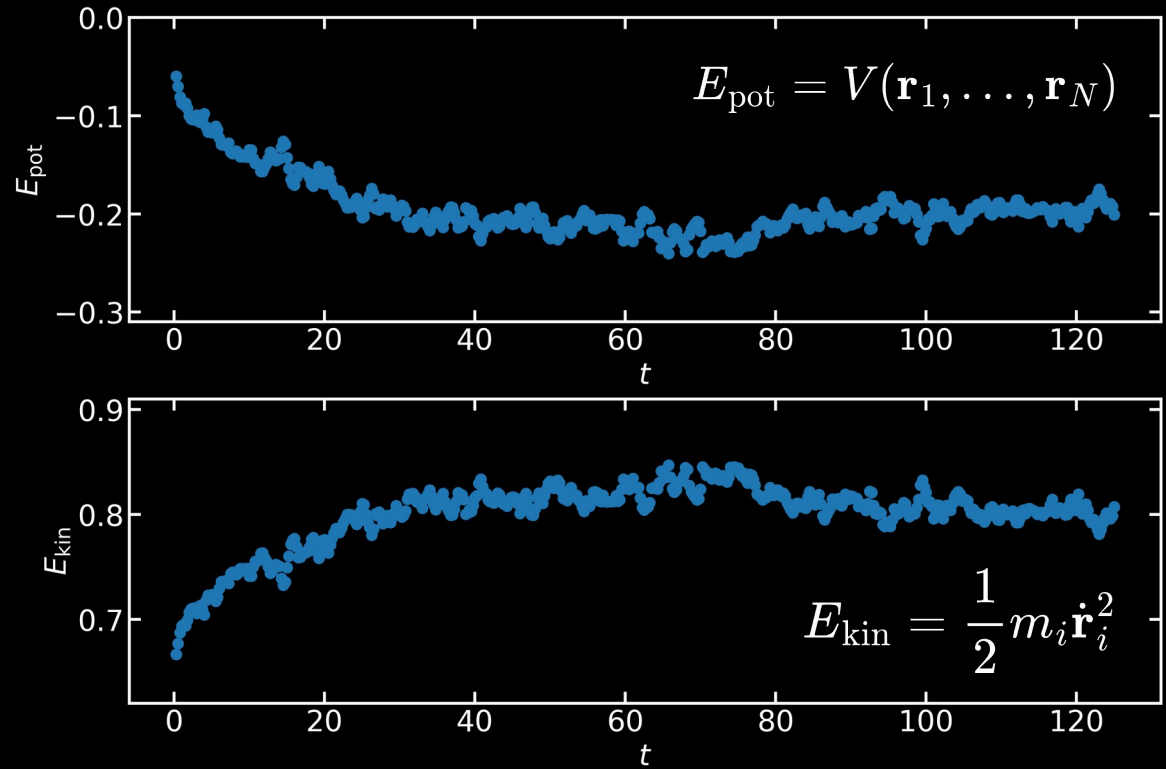
- 40 x 40 x 40 box
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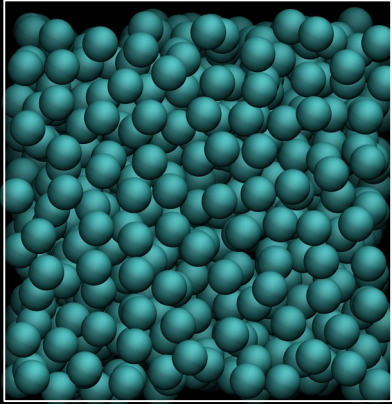
Energies from MD



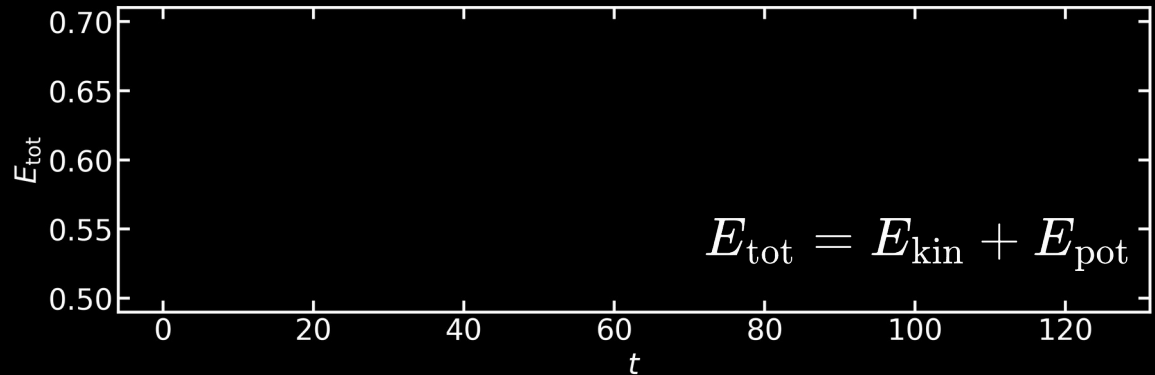
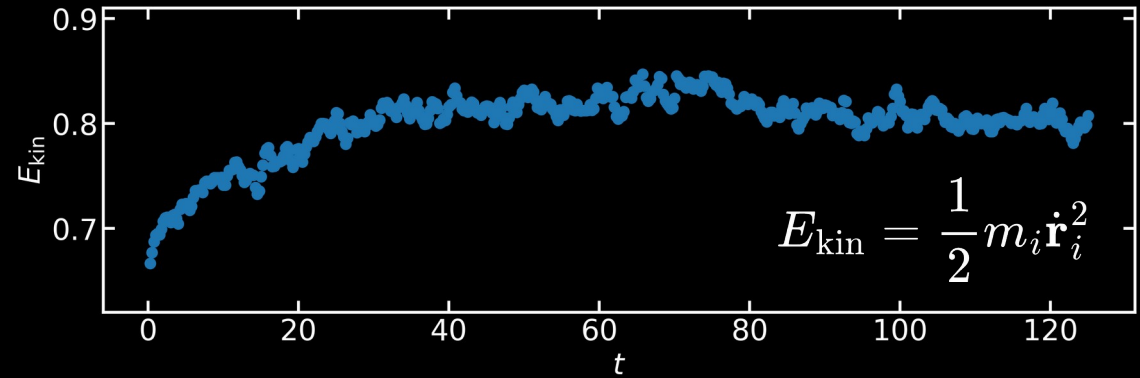
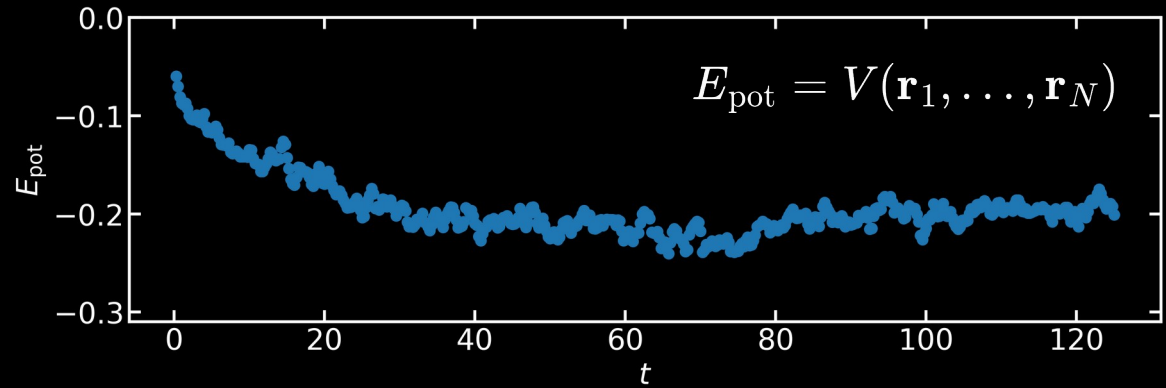
- 40 x 40 x 40 box
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- Define LJ potential
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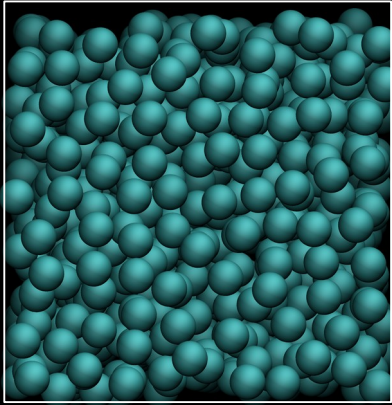
Energies from MD



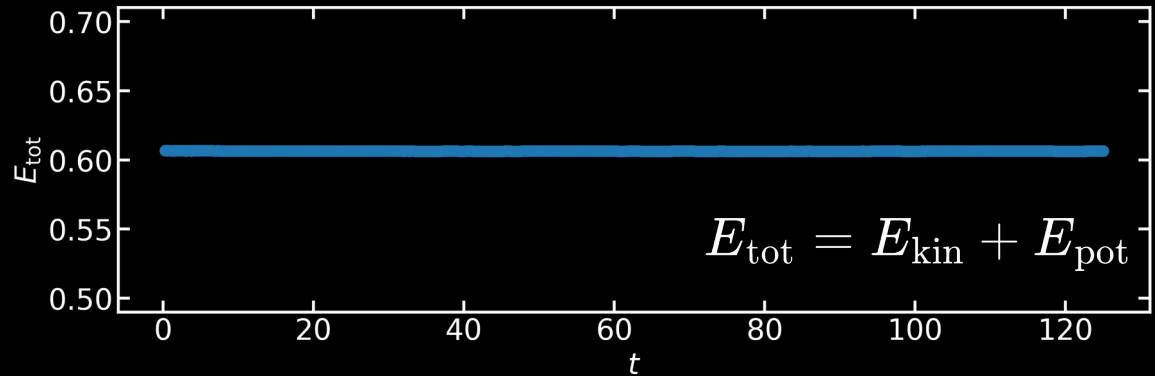
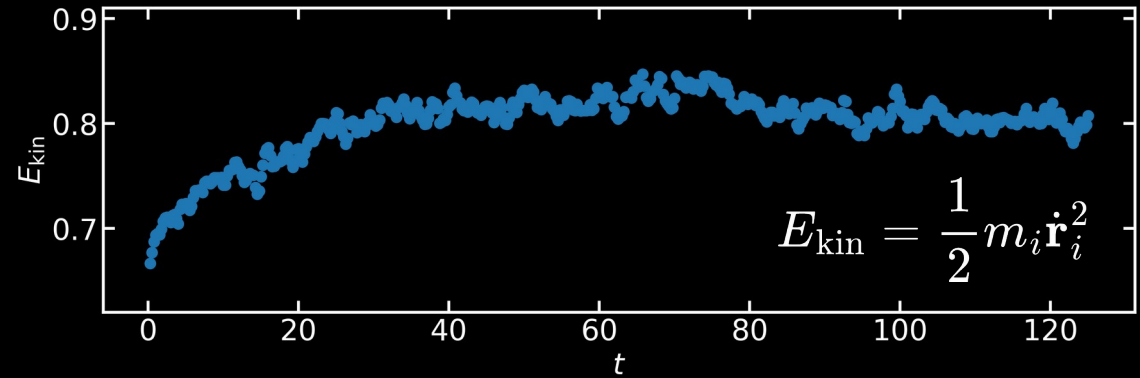
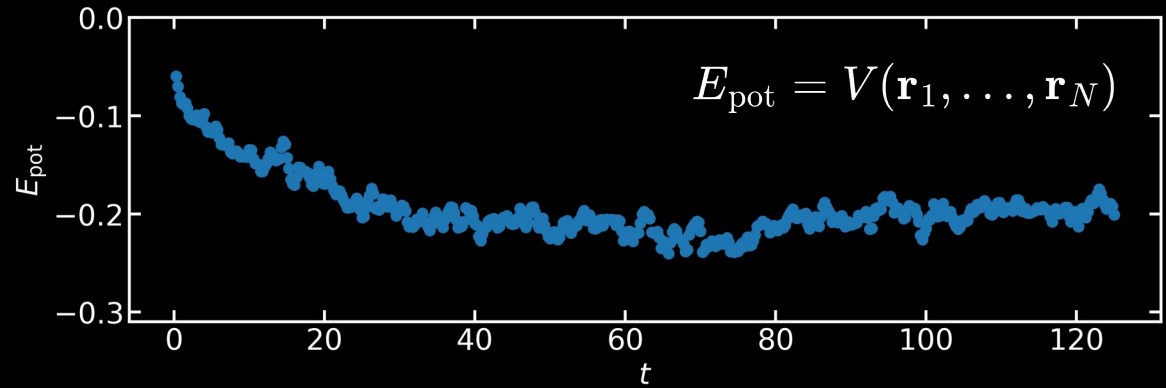
- 40 x 40 x 40 box
- 600 atoms
- Define LJ potential
- Apply Verlet algorithm



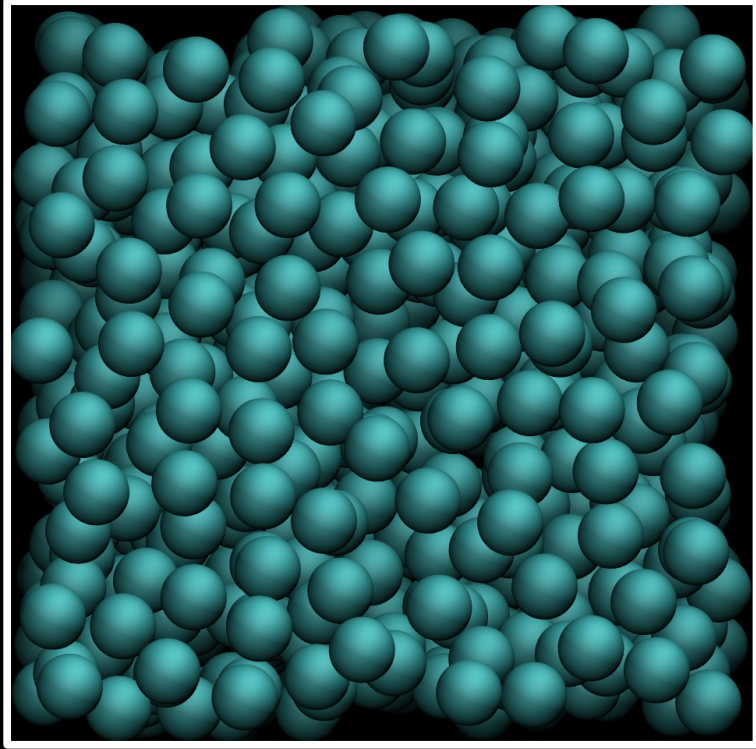
Energies from MD



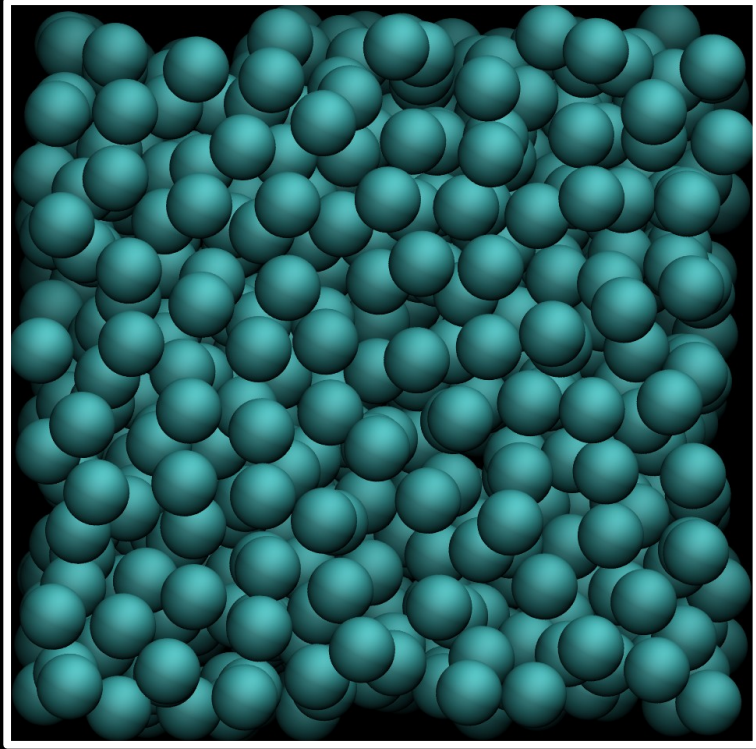
- 40 x 40 x 40 box
- 600 atoms
- Define LJ potential
- Apply Verlet algorithm



Thermodynamic ensembles

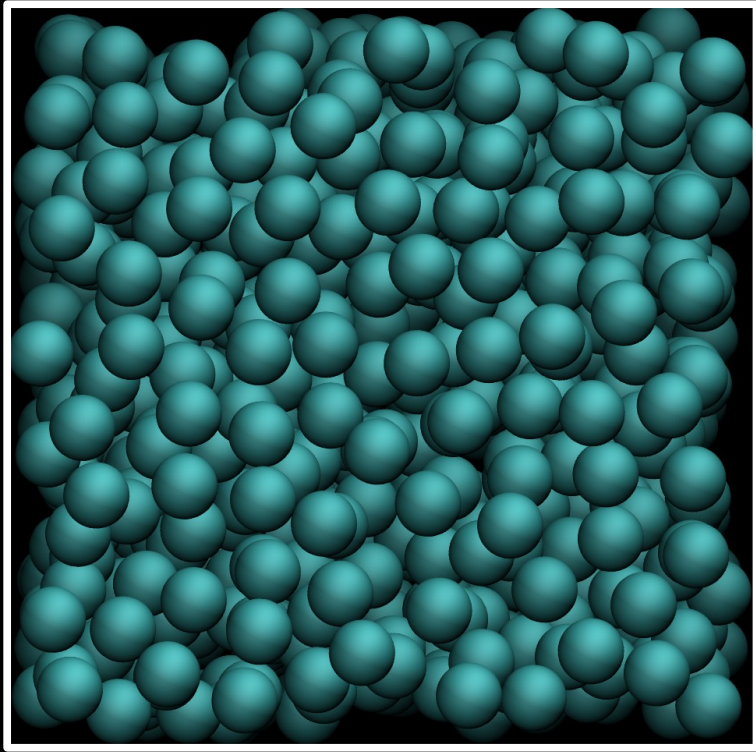


Thermodynamic ensembles



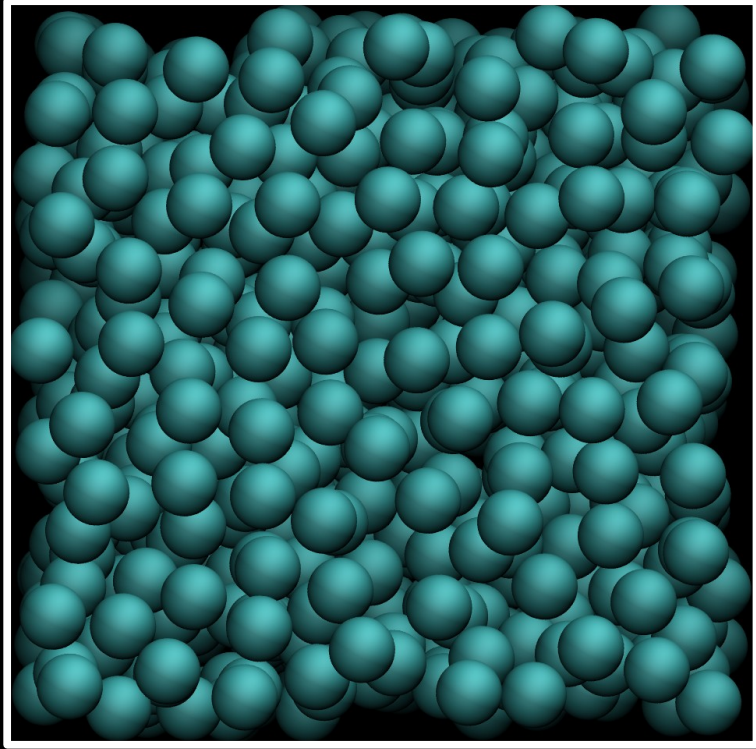
- fixed particle number (N)

Thermodynamic ensembles



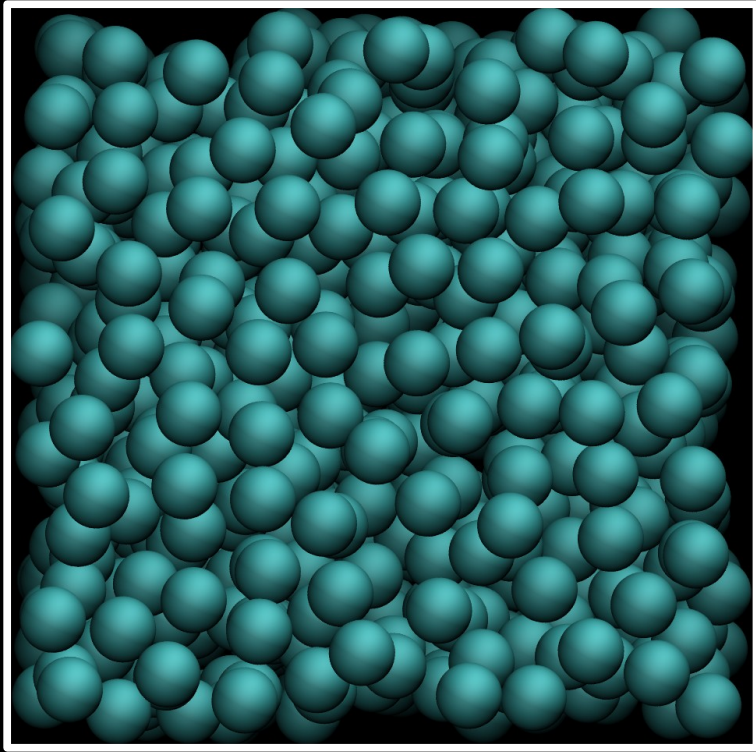
- fixed particle number (N)
- fixed box volume (V)

Thermodynamic ensembles



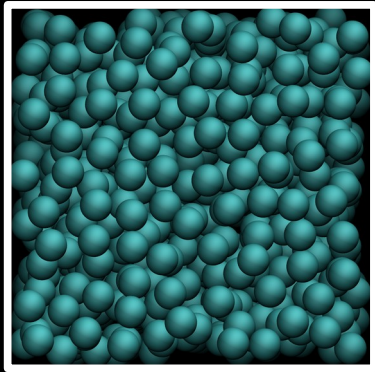
- fixed particle number (N)
- fixed box volume (V)
- constant energy (E)

Thermodynamic ensembles



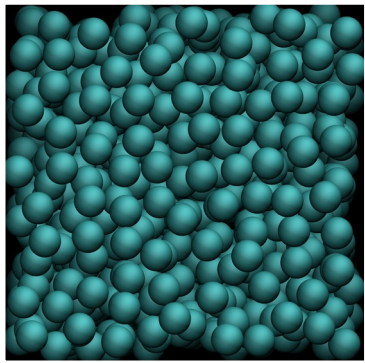
- fixed particle number (N)
 - fixed box volume (V)
 - constant energy (E)
- Microcanonical (NVE)

Thermostats : Mimic energy exchange with a bath



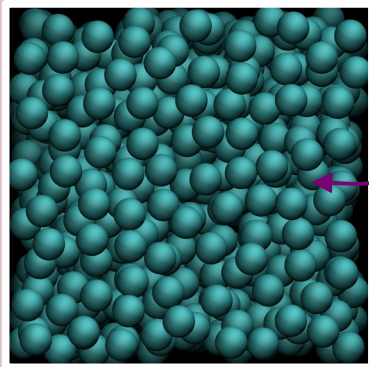
Thermostats : Mimic energy exchange with a bath

Bath with
temperature T_0



Thermostats : Mimic energy exchange with a bath

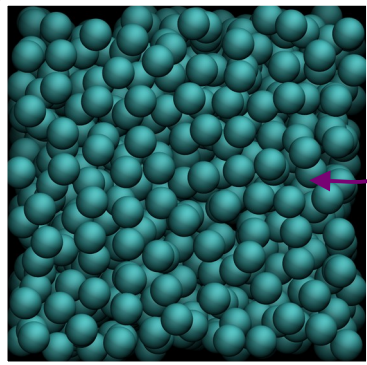
Bath with
temperature T_0



Energy

Thermostats : Mimic energy exchange with a bath

Bath with
temperature T_0



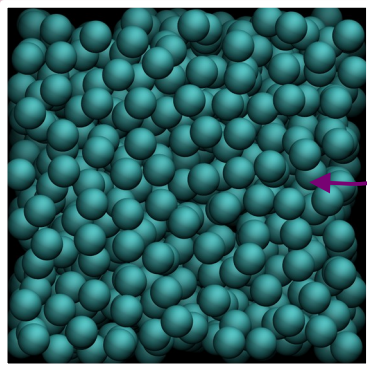
Energy

Simple thermostat: rescale velocities

$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

Thermostats : Mimic energy exchange with a bath

Bath with
temperature T_0



Energy

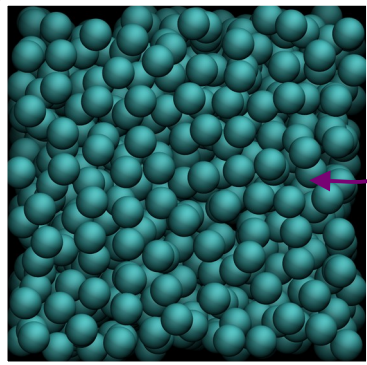
Simple thermostat: rescale velocities

$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left(\frac{T_0}{T} - 1 \right)}$$

Thermostats : Mimic energy exchange with a bath

Bath with
temperature T_0



Energy

Simple thermostat: rescale velocities

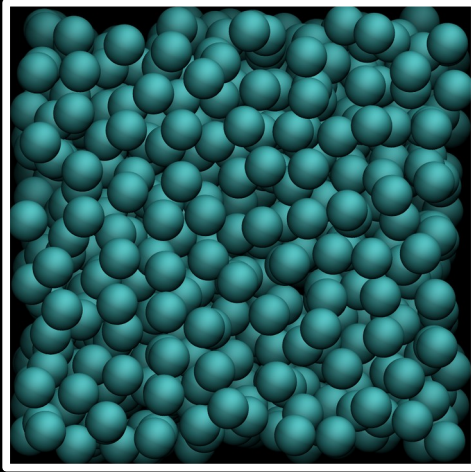
$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

Target temperature

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left(\frac{T_0}{T} - 1 \right)}$$

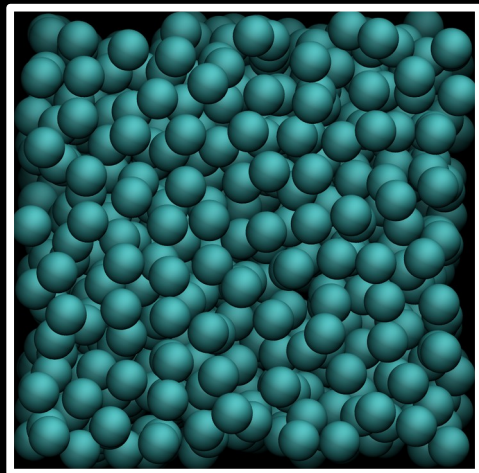
Coupling time

Effect thermostats

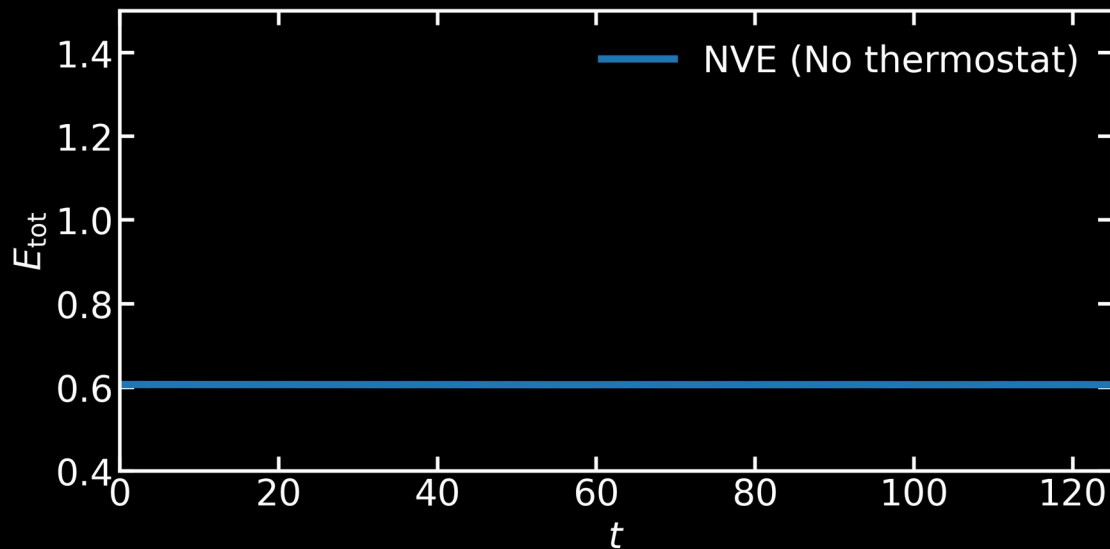
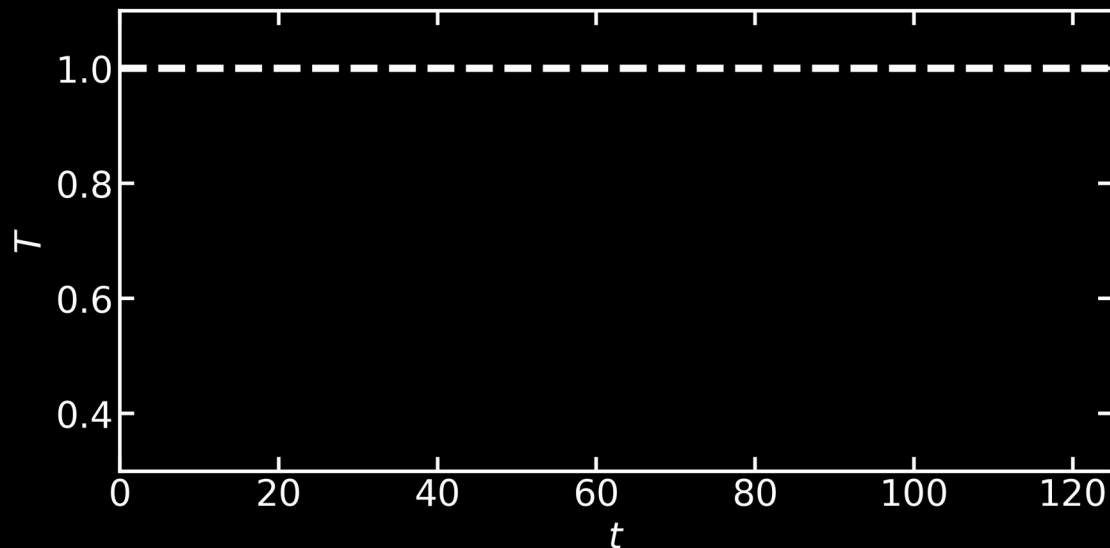


- Create a $40 \times 40 \times 40$ box
- Place 600 atoms into it
- Use LJ potential
- Apply Verlet algorithm

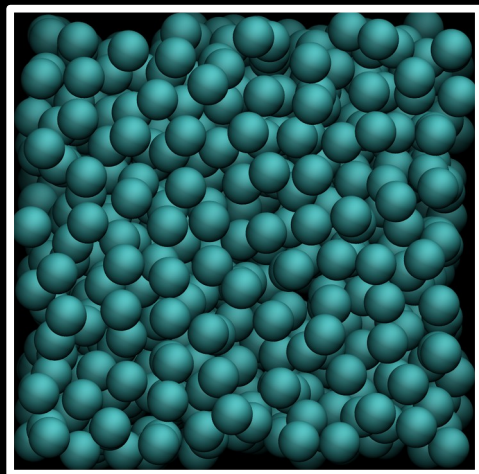
Effect thermostats



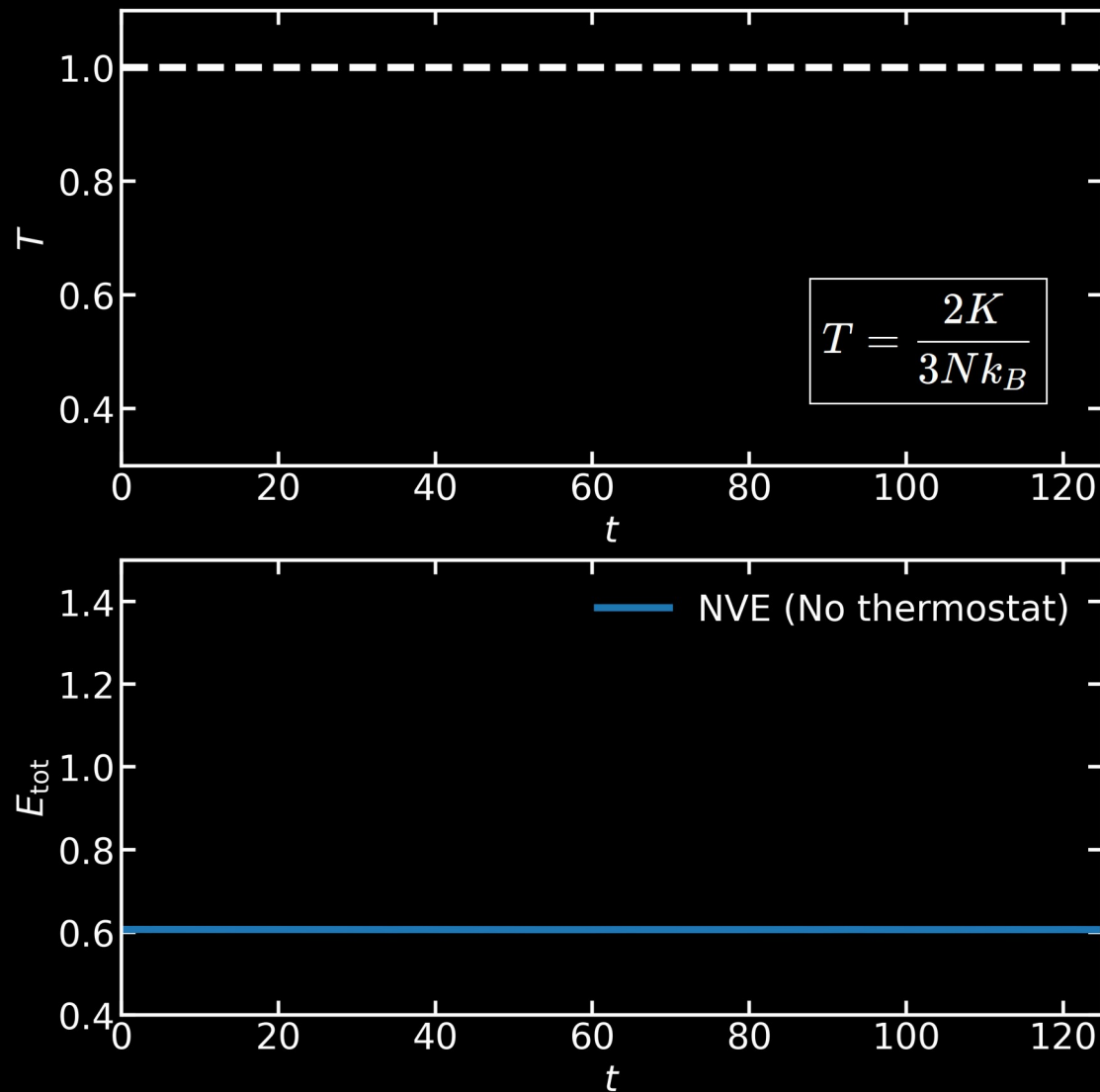
- Create a $40 \times 40 \times 40$ box
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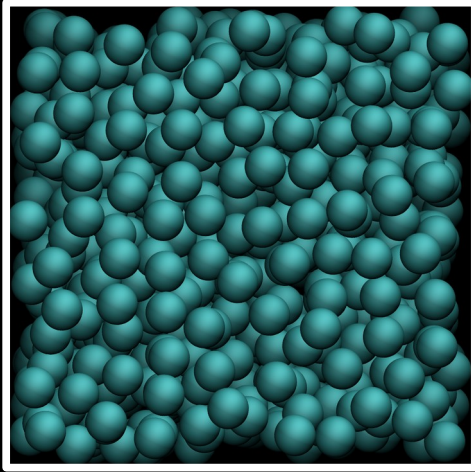
Effect thermostats



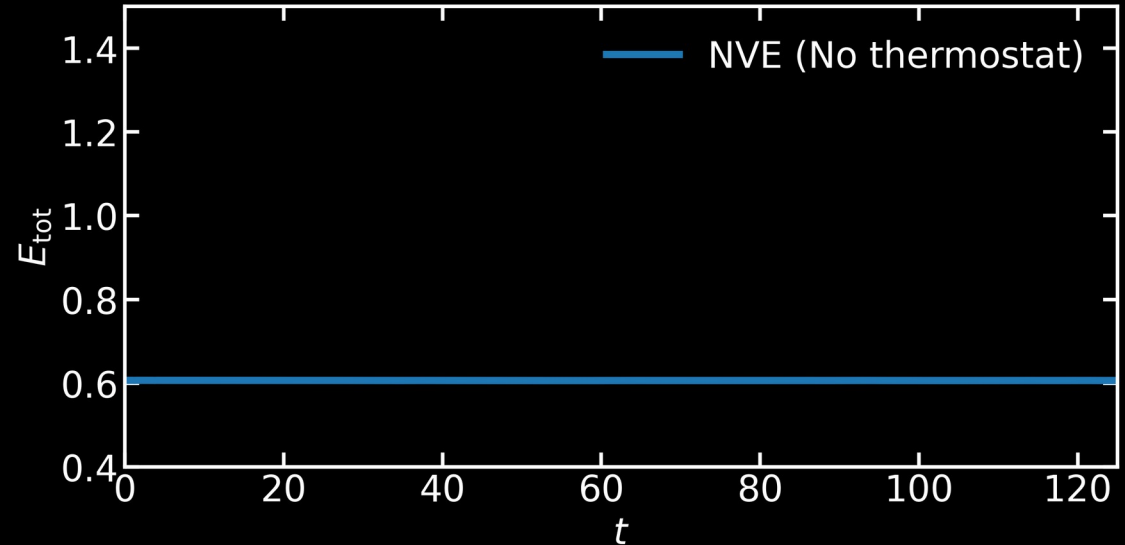
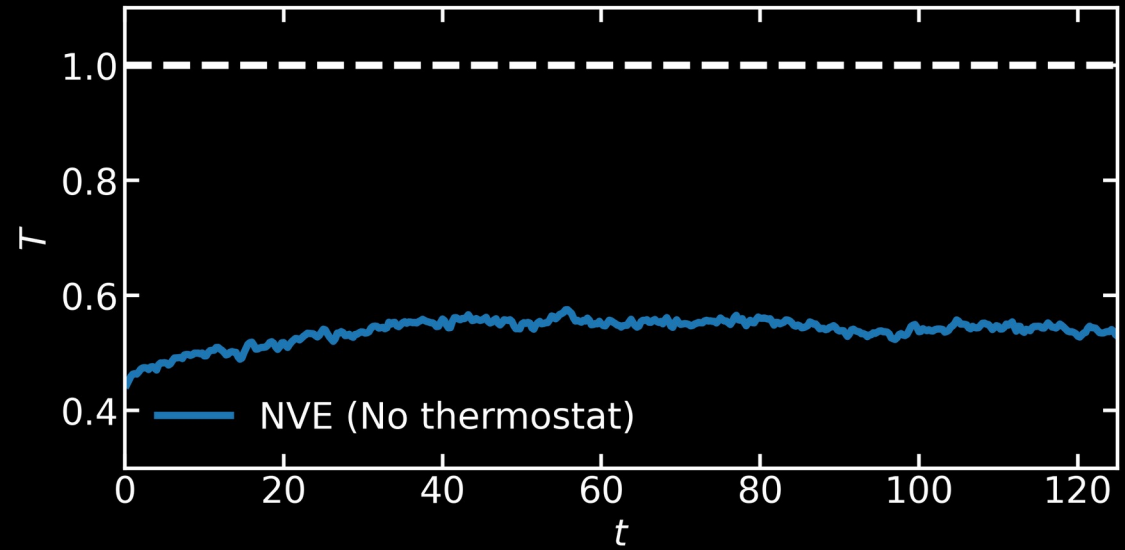
- Create a 40 x 40 x 40 box
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- Apply Verlet algorithm



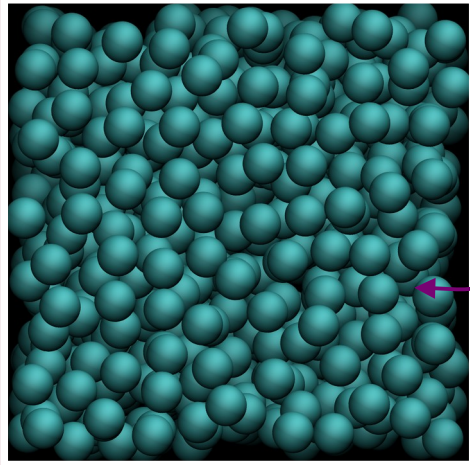
Effect thermostats



- Create a $40 \times 40 \times 40$ box
- Place 600 atoms into it
- Use LJ potential
- Apply Verlet algorithm



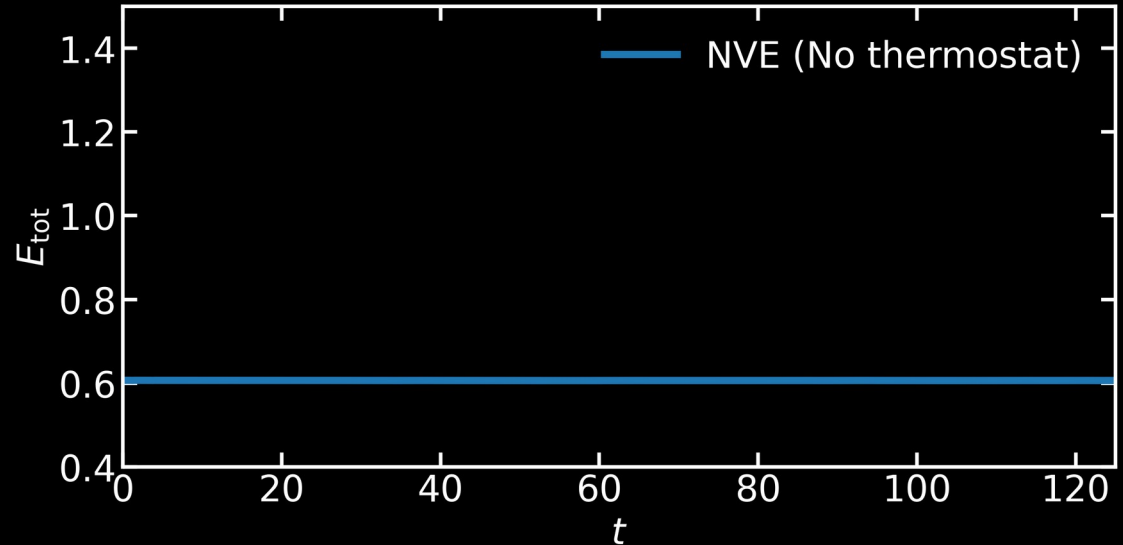
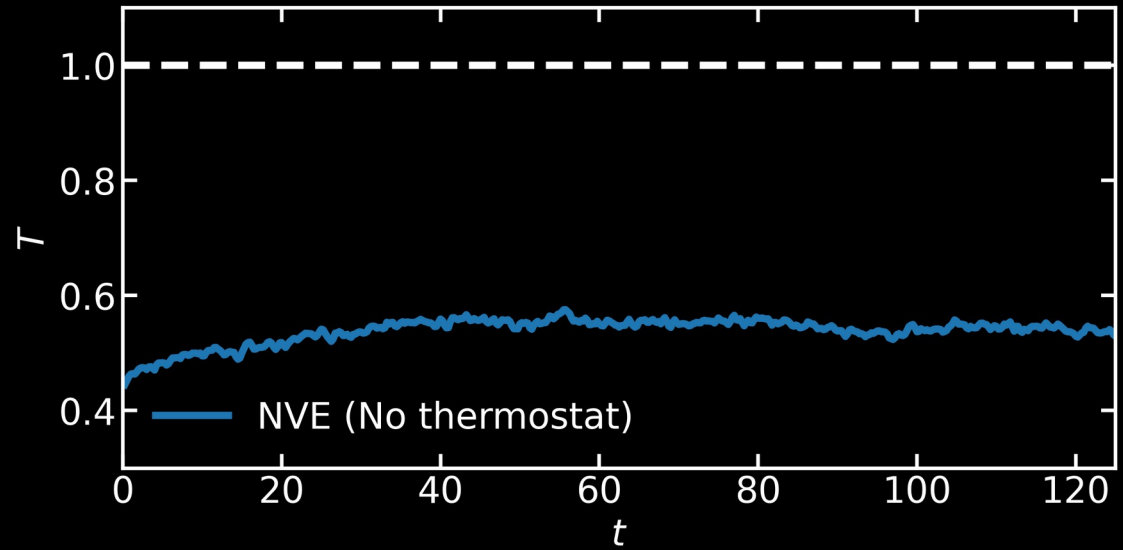
Effect thermostats



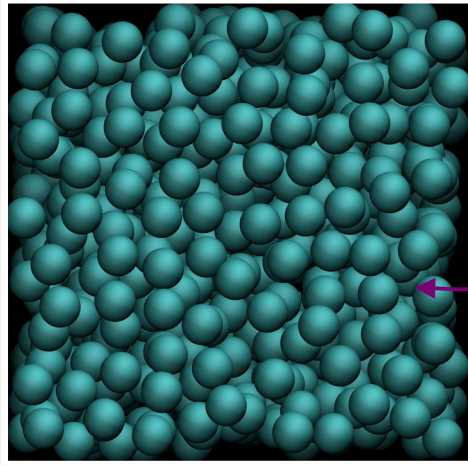
$$T_0 = 1$$

Energy

- Create a $40 \times 40 \times 40$ box
- Place 600 atoms into it
- Use LJ potential
- Apply Verlet algorithm
- Apply thermostat ($T_0 = 1$)



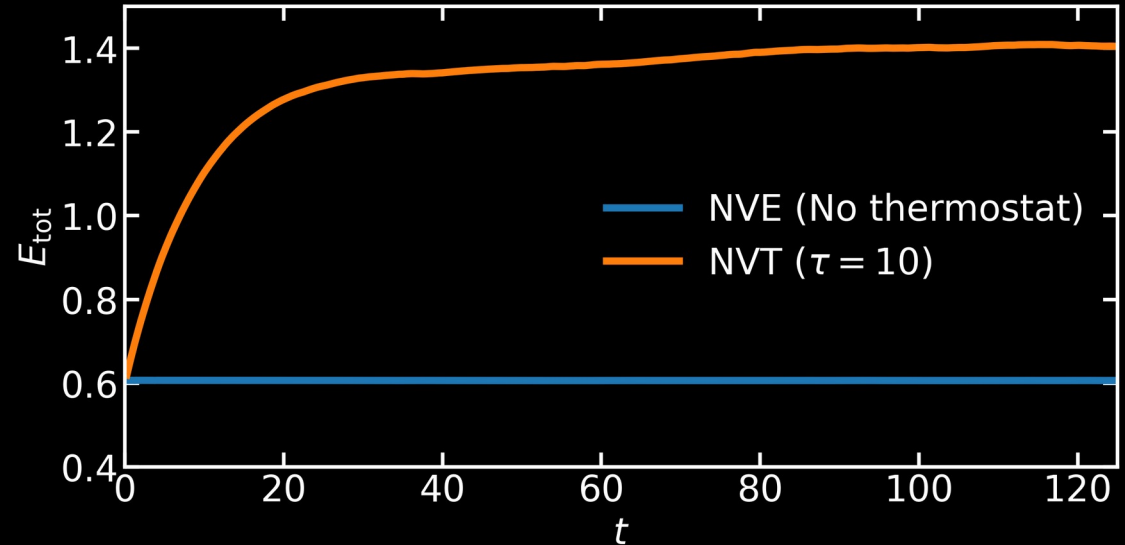
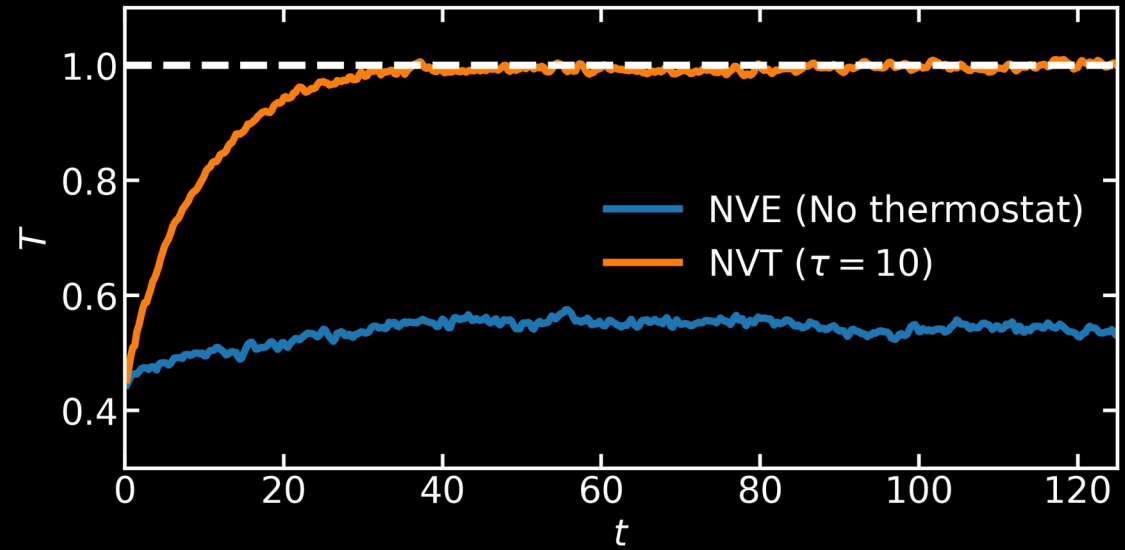
Effect thermostats



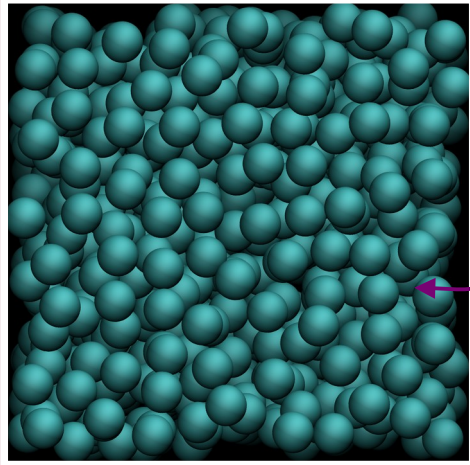
$$T_0 = 1$$

Energy

- Create a $40 \times 40 \times 40$ box
- Place 600 atoms into it
- Use LJ potential
- Apply Verlet algorithm
- Apply thermostat ($T_0 = 1$)



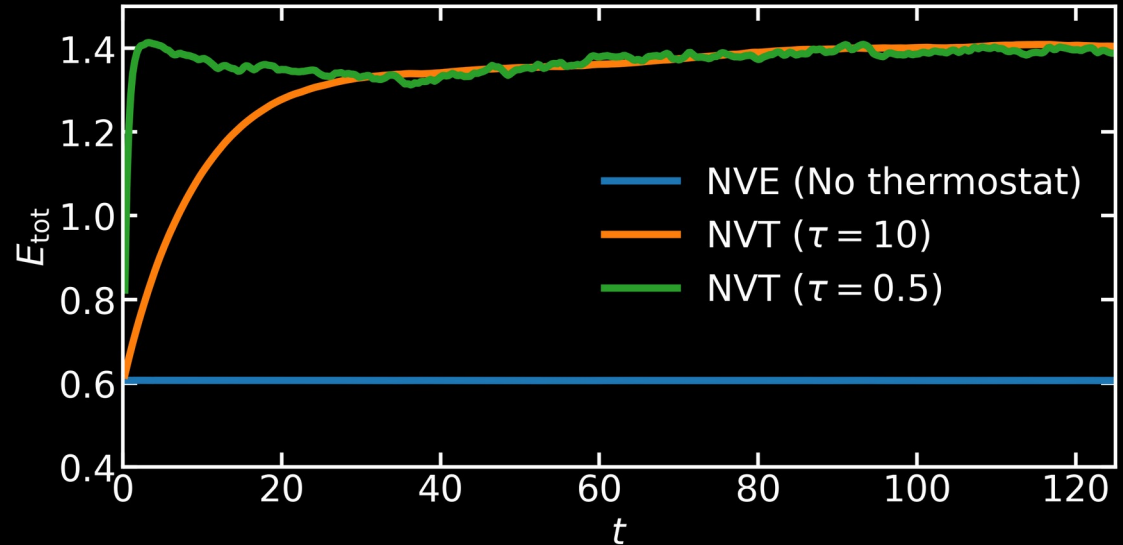
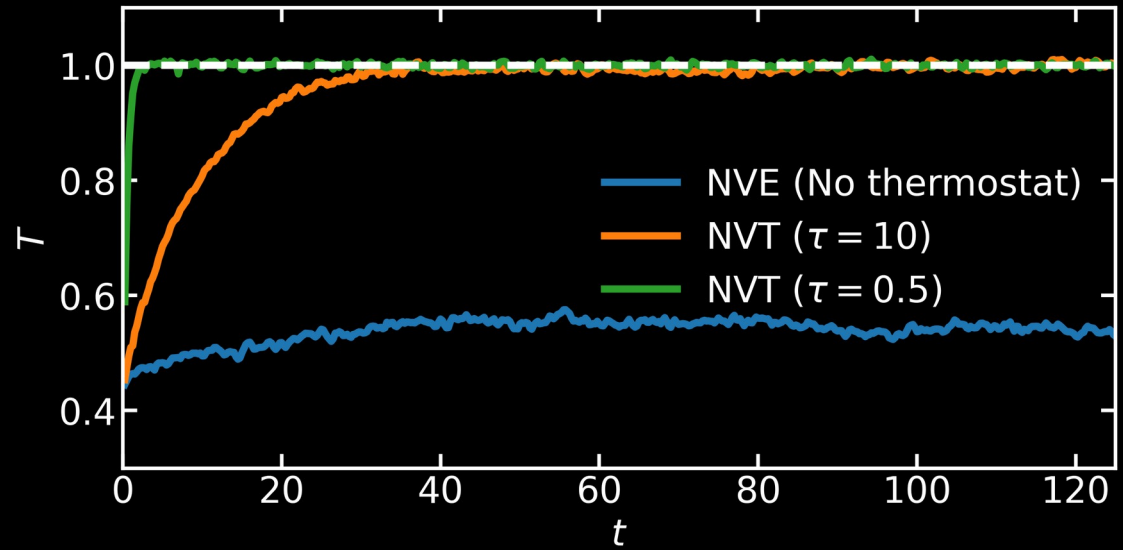
Effect thermostats



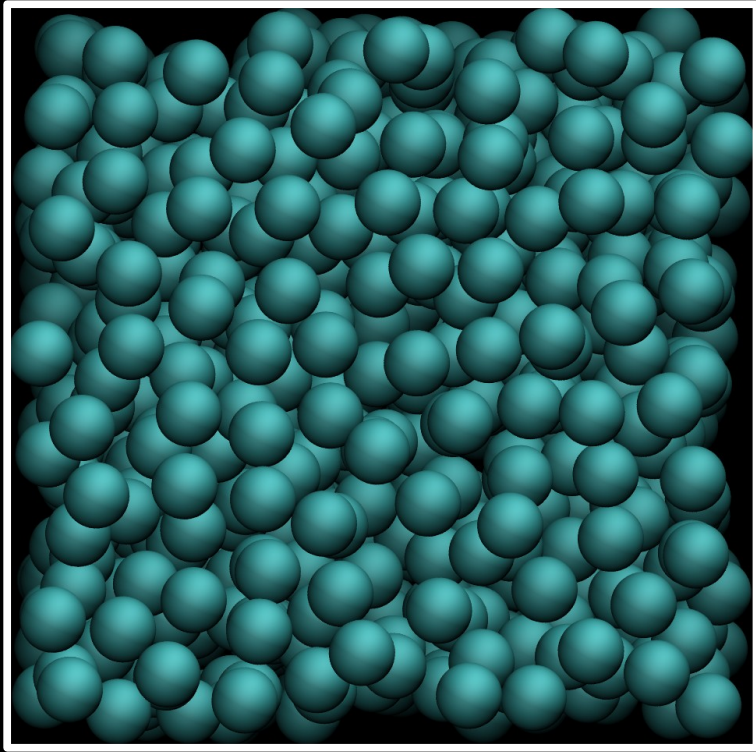
$$T_0 = 1$$

Energy

- Create a $40 \times 40 \times 40$ box
- Place 600 atoms into it
- Use LJ potential
- Apply Verlet algorithm
- Apply thermostat ($T_0 = 1$)



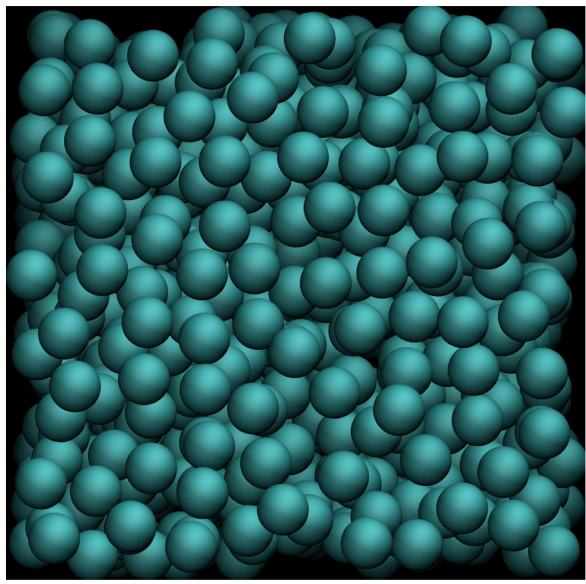
Thermodynamic ensembles



- fixed particle number (N)
 - fixed box volume (V)
 - constant energy (E)
- Microcanonical (NVE)

Thermodynamic ensembles

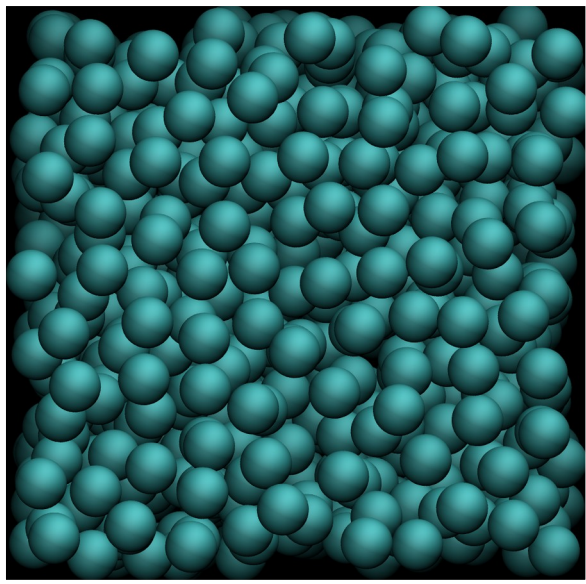
Bath (T_0)



- fixed particle number (N)
 - fixed box volume (V)
 - constant energy (E)
- Microcanonical (NVE)

Thermodynamic ensembles

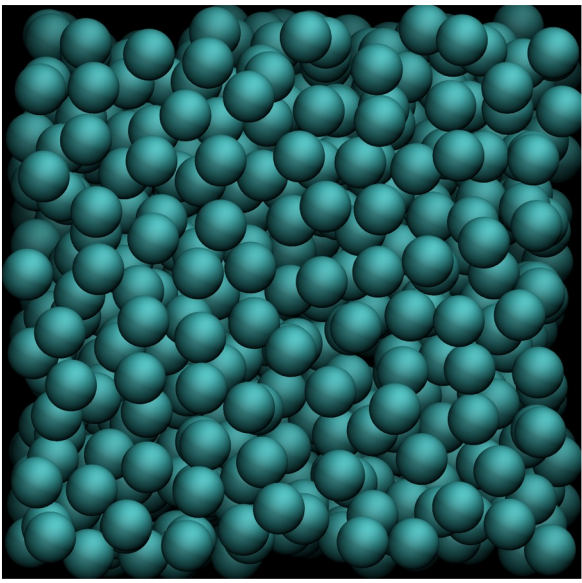
Bath (T_0)



- fixed particle number (N)
- fixed box volume (V)
- ~~- constant energy (E)~~
- ~~→ Microcanonical (NVE)~~

Thermodynamic ensembles

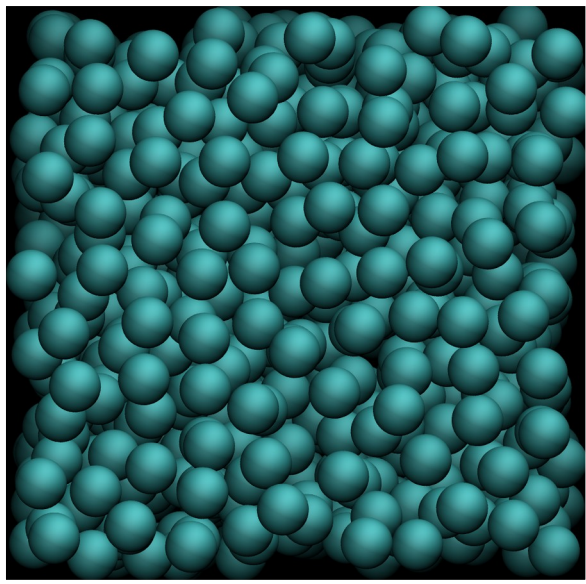
Bath (T_0)



- fixed particle number (N)
- fixed box volume (V)
- constant temperature (T)

Thermodynamic ensembles

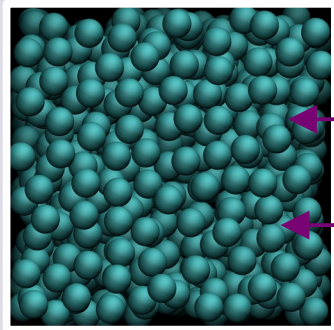
Bath (T_0)



- fixed particle number (N)
 - fixed box volume (V)
 - constant temperature (T)
- Canonical (NVT)

Barostat : Mimics the exchange of volume with a bath

Bath with temperature T_0 and pressure P_0)



Energy

Volume

Simple barostat: rescale box volume

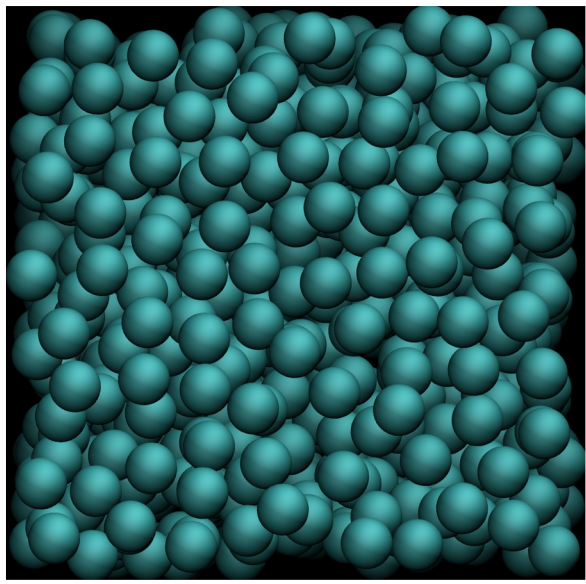
$$\mathbf{V}_i \rightarrow \mu \mathbf{V}_i$$

$$\mu = 1 - \frac{\Delta t}{\tau} \kappa_T (P_0 - P)$$

↖ Isothermal compressibility

Thermodynamic ensembles

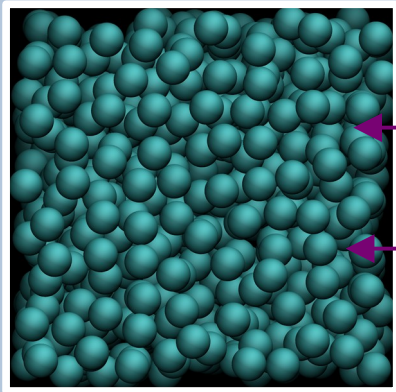
Bath (T_0, P_0)



- fixed particle number (N)
 - constant pressure (P)
 - constant temperature (T)
- Isothermal-isobaric (NPT)

Thermodynamic ensembles

Bath (T_0, μ_0)

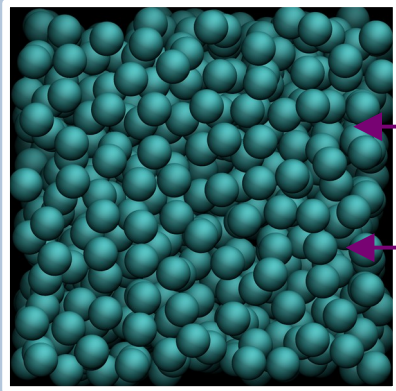


Energy

Particles

Thermodynamic ensembles

Bath (T_0, μ_0)



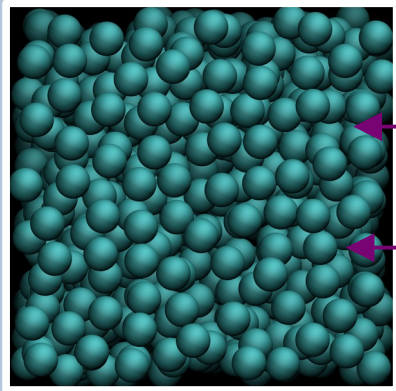
Energy

Particles

- constant chemical potential (μ)
- constant volume (V)
- constant temperature (T)

Thermodynamic ensembles

Bath (T_0, μ_0)



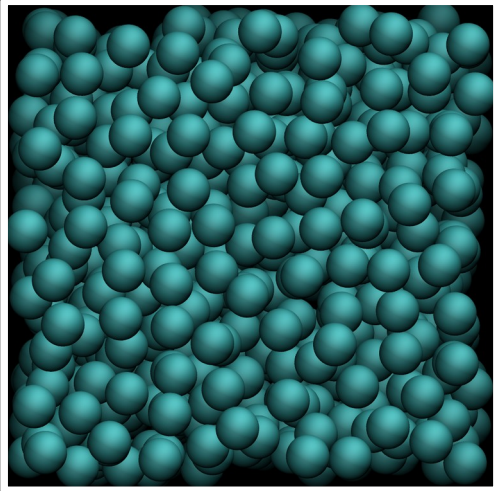
Energy

Particles

- constant chemical potential (μ)
 - constant volume (V)
 - constant temperature (T)
- Grand canonical (μVT)

Molecular dynamics

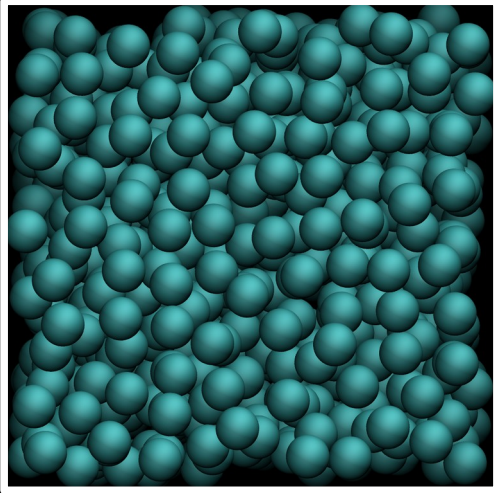
Periodic boundary conditions



- 1) Define the system
- 2) Measure forces between atoms $\mathbf{F}_{ij}$
- 3) Solve Newton's equations of motion
- 4) Move atoms $\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$
- 5) Optional : Apply thermostat or barostat

Molecular dynamics

Periodic boundary conditions

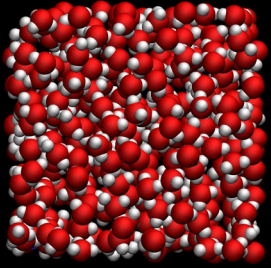


- 1) Define the system
- 2) Measure forces between atoms $\mathbf{F}_{ij}$
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- 5) Optional : Apply thermostat or barostat

Good model for argon or other simple fluids

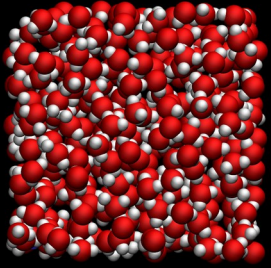
Simulating systems more complex than argon

Simulating systems more complex than argon

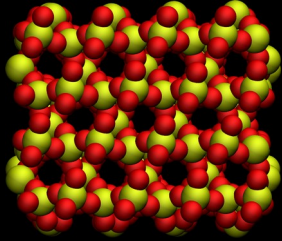


1) Ions, molecules, proteins, etc

Simulating systems more complex than argon

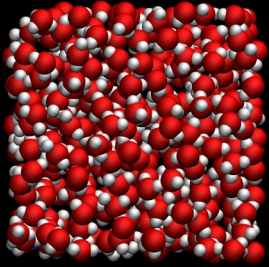


1) Ions, molecules, proteins, etc

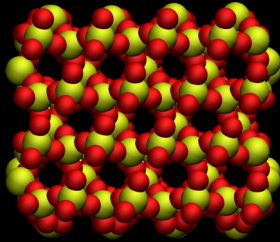


2) Metals and covalent materials

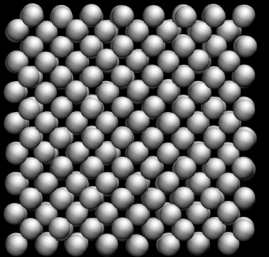
Simulating systems more complex than argon



1) Ions, molecules, proteins, etc

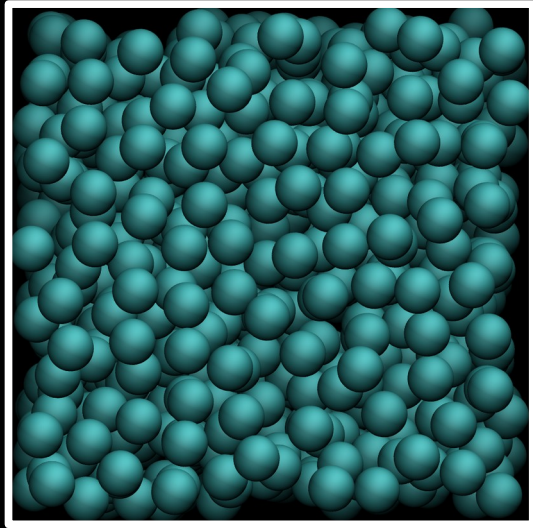


2) Metals and covalent materials

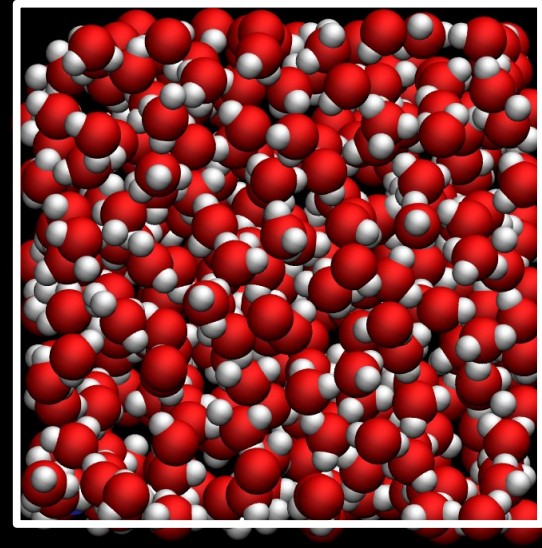


3) About MLPI

Simulating ions, molecules, proteins, ...

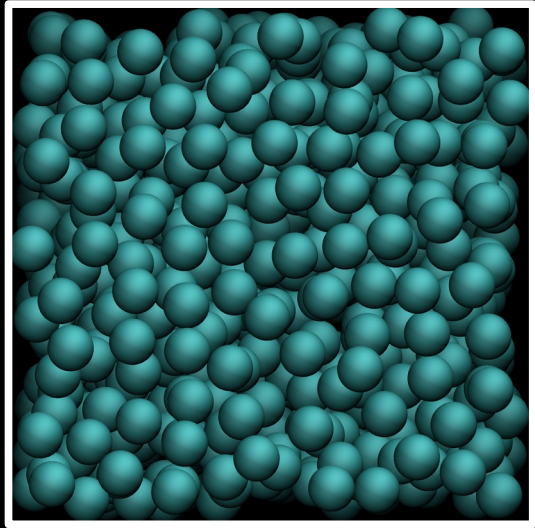


Simple fluid

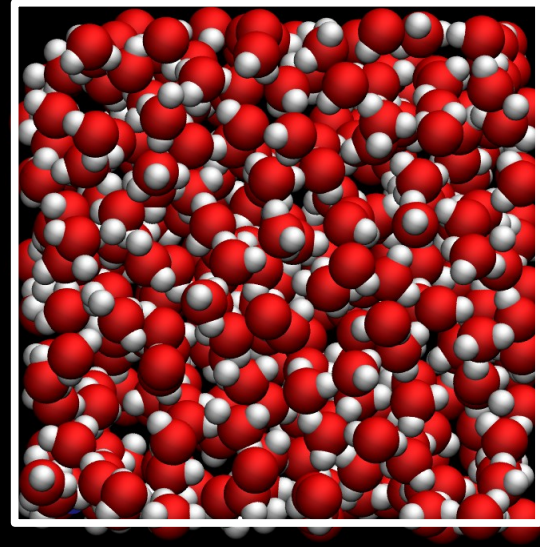


Water

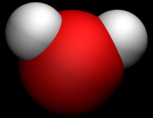
Simulating ions, molecules, proteins, ...



Simple fluid

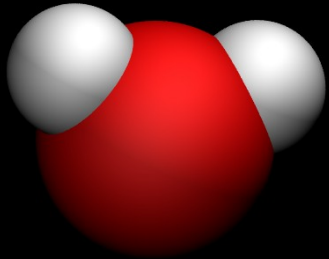


Water



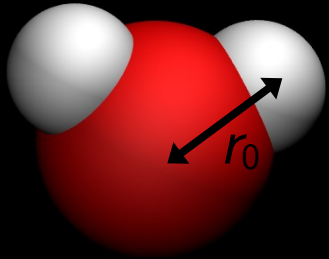
Simulating ions, molecules, proteins, ...

1) Connect atoms by bonds



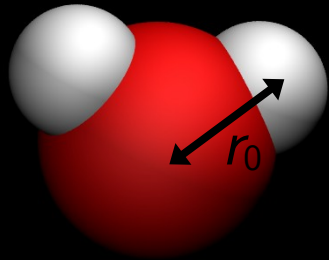
Simulating ions, molecules, proteins, ...

1) Connect atoms by bonds



Simulating ions, molecules, proteins, ...

1) Connect atoms by bonds

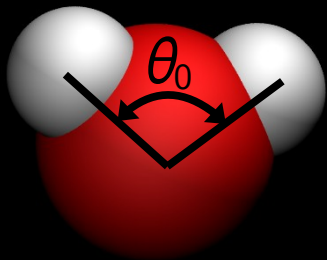


$$U_{\text{bond}}(r) = \frac{1}{2}k(r - r_0)^2$$

bond stiffness

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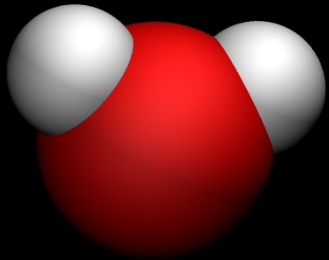
$$U_{\text{angle}}(\theta) = \frac{1}{2}k_{\theta}(\theta - \theta_0)^2$$

angle bending constant

Simulating ions, molecules, proteins, ...

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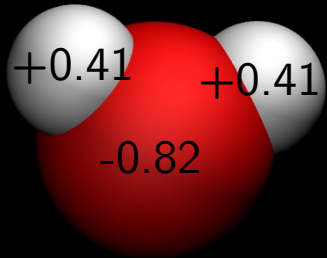
2) Attribute partial charges to atoms



Simulating ions, molecules, proteins, ...

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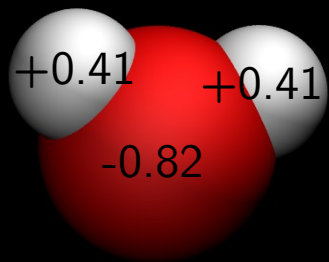
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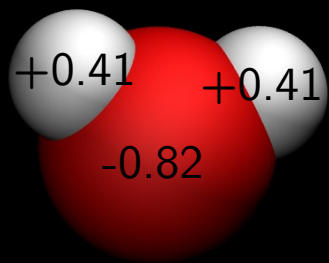


- Represent average redistribution of electron density
- Help reproduce the molecule's electrostatic behavior

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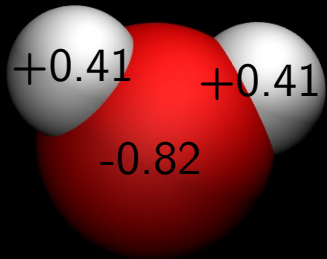
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- Additional potential :

$$U_{\text{elec}} = \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}}$$

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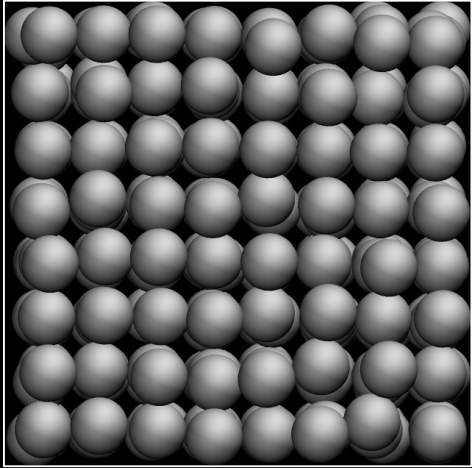


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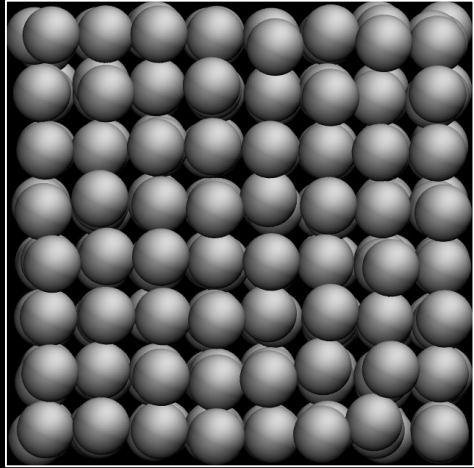
→ long ranged
(Ewald-based methods)

Simulating metals



Al (fcc)

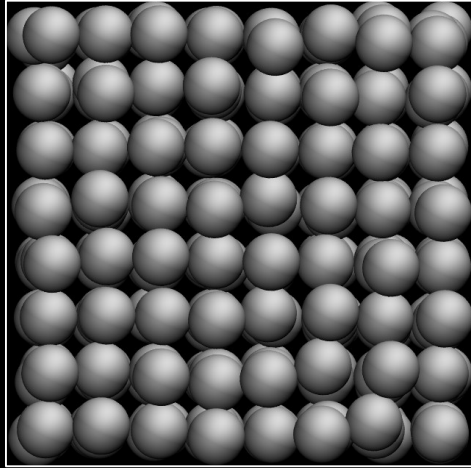
Simulating metals



Al (fcc)

Metals : Bonding depends on the local electronic environment (not just on pairwise distances)

Simulating metals



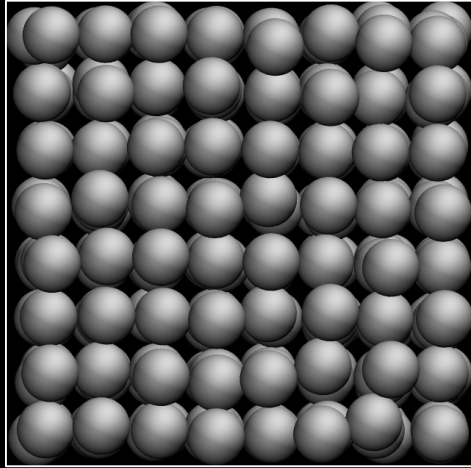
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Embedded atom model (EAM)

$$U_i = F_i(\rho_i) + \frac{1}{2} \sum_{j \neq i} \phi_{ij}(r_{ij})$$

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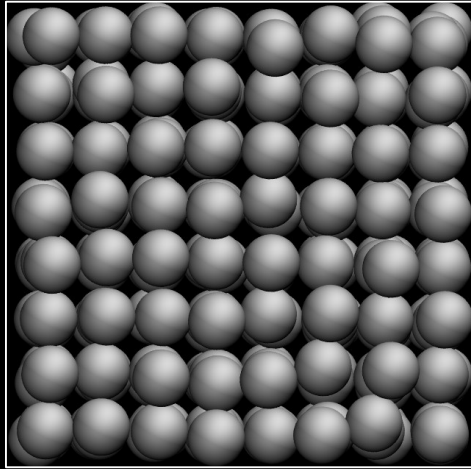
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Pairwise function

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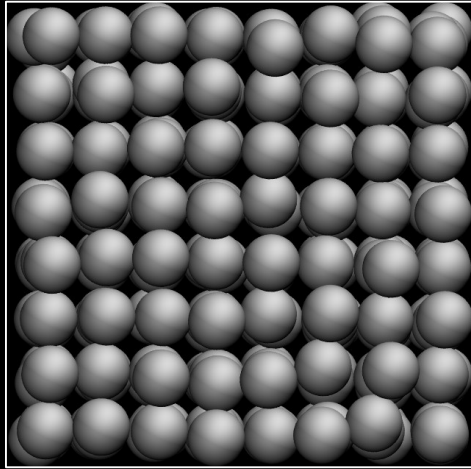
Embedded atom model (EAM)

$$U_i = F_i(\rho_i) + \frac{1}{2} \sum_{j \neq i} \phi_{ij}(r_{ij})$$

Many-body term : Embedding energy
(function of local electron density ρ_i)

Pairwise function

Simulating metals



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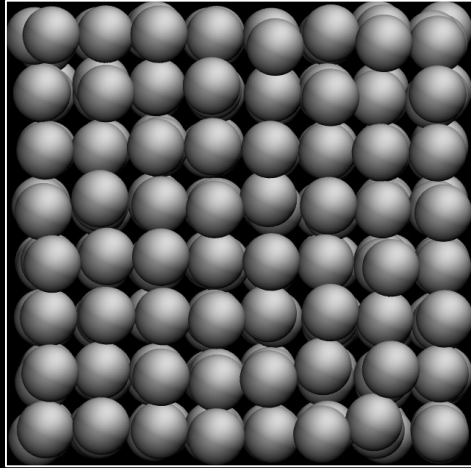
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$$\rho_i = \sum_{j \neq i} f_j(r_{ij})$$

Electron-density contribution
from atom j

Simulating metals



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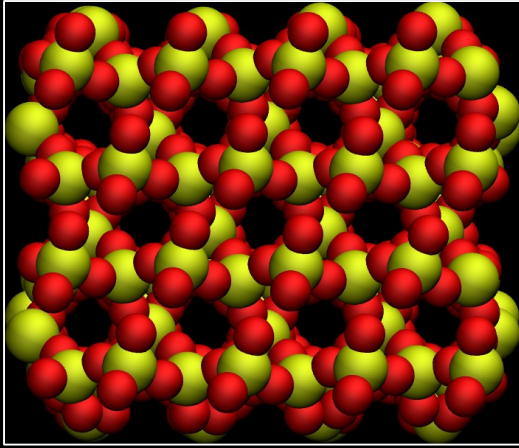
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Embedded atom model (EAM)

Many-body interactions

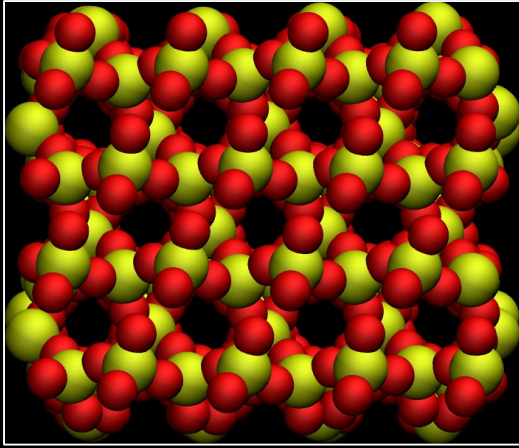
→ coordination, crystal structure, surface, defects

Simulating covalent materials



SiO₂ (Quartz)

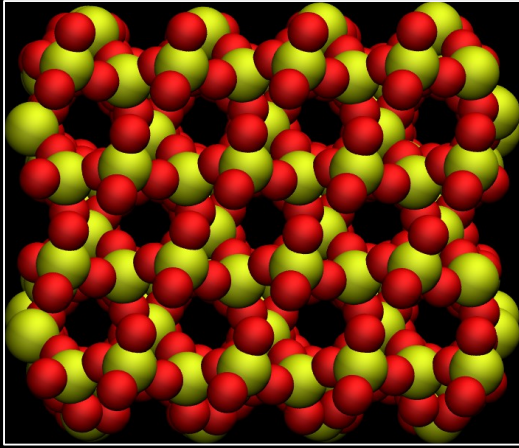
Simulating covalent materials



SiO₂ (Quartz)

Covalent materials : atoms are held together by covalent chemical bonds that are directional

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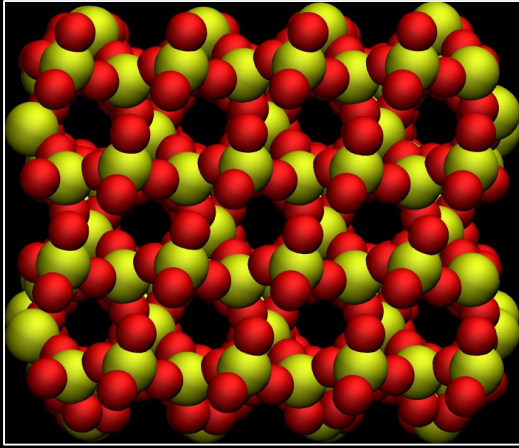
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Tersoff

$$E = \frac{1}{2} \sum_i \sum_{j \neq i} f_c(r_{ij}) [f_R(r_{ij}) + b_{ij} f_A(r_{ij})]$$

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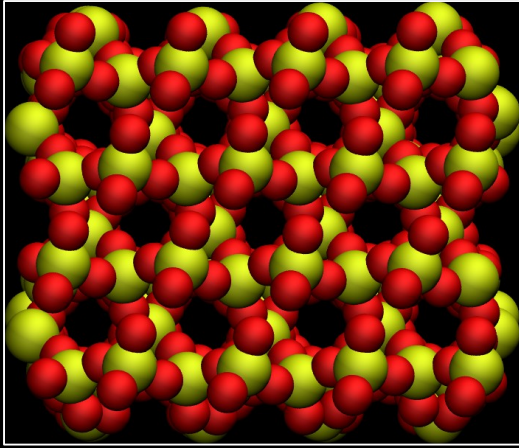
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Attractive term

Bond-order term

Repulsive pairwise term

Simulating covalent materials



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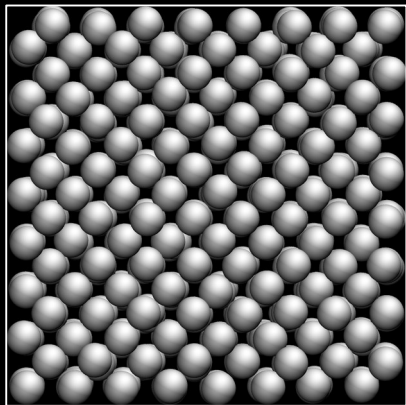
$$E = \frac{1}{2} \sum_i \sum_{j \neq i} f_c(r_{ij}) [f_R(r_{ij}) + b_{ij} f_A(r_{ij})]$$

$$b_{ij} = (1 + \beta^n \zeta_{ij}^n)^{-1/(2n)}$$

$$\zeta_{ij} = \sum_{k \neq i, j} f_c(r_{ik}) g(\theta_{ijk})$$

Angle between 3 atoms

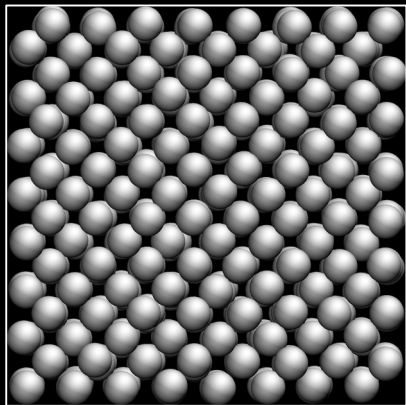
Machine-learning interatomic potentials MLPI



Ge (diamond)

No fixed functional form

Machine-learning interatomic potentials MLPI

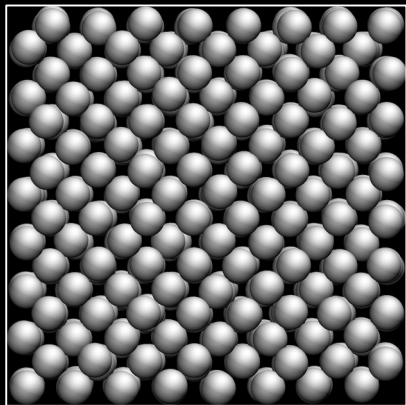


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Learns the energy landscape from quantum-mechanical data

Machine-learning interatomic potentials MLPI



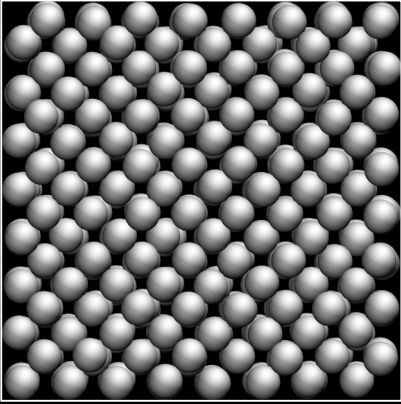
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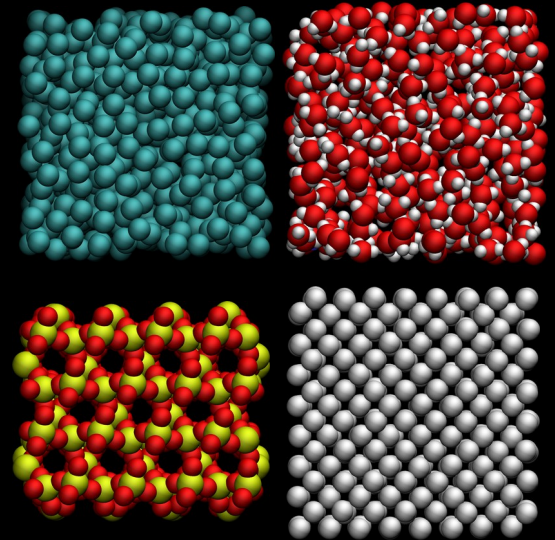
From the LAMMPS user perspective : as easy to use as EAM or Tersoff

Takeaway message : MD

- 1) Define the system

Takeaway message : MD

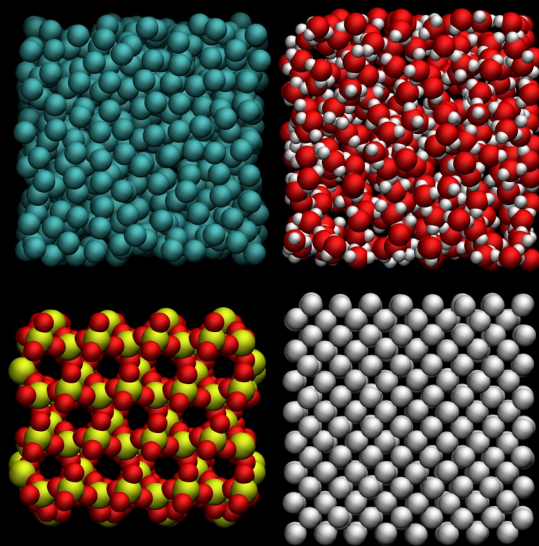
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Takeaway message : MD

1) Define the system

2) Measure forces between atoms $F_{ij} = -\frac{dU(r_{ij})}{dr_{ij}}$



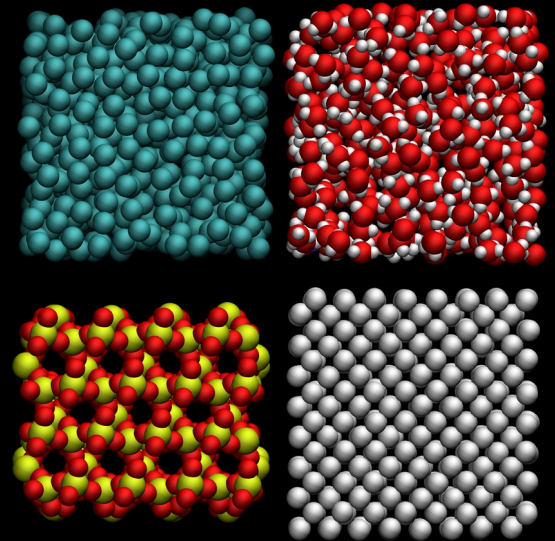
Takeaway message : MD

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$$U_{ij}(r_{ij}) = 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \quad (\text{Lennard-Jones})$$

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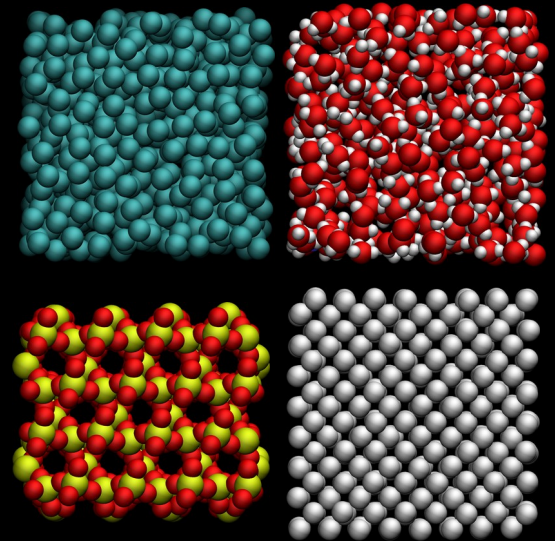
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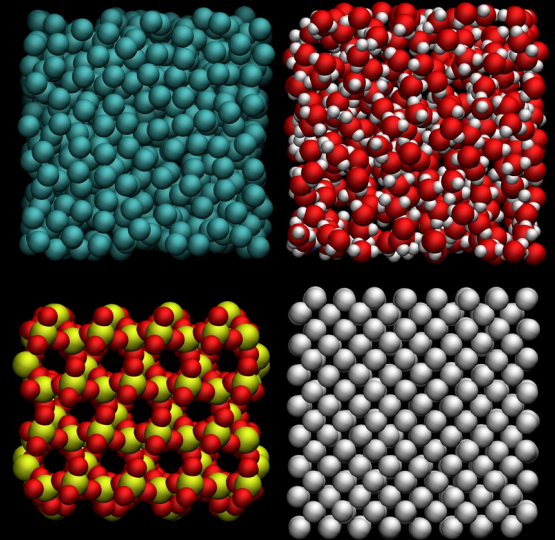
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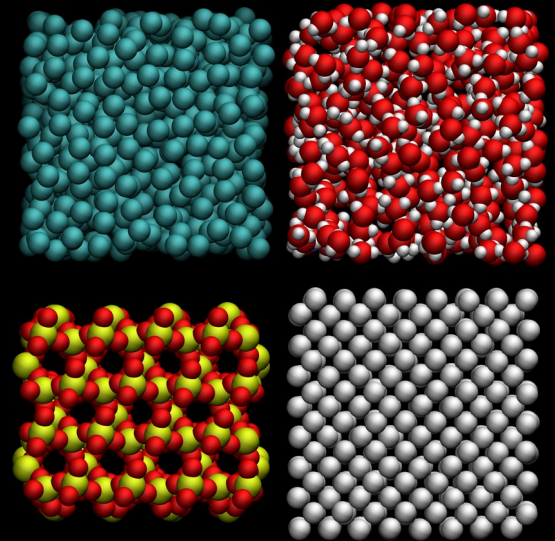
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$\rightarrow$ Extract observables from trajectory (energies, structure, dynamics, etc.)

